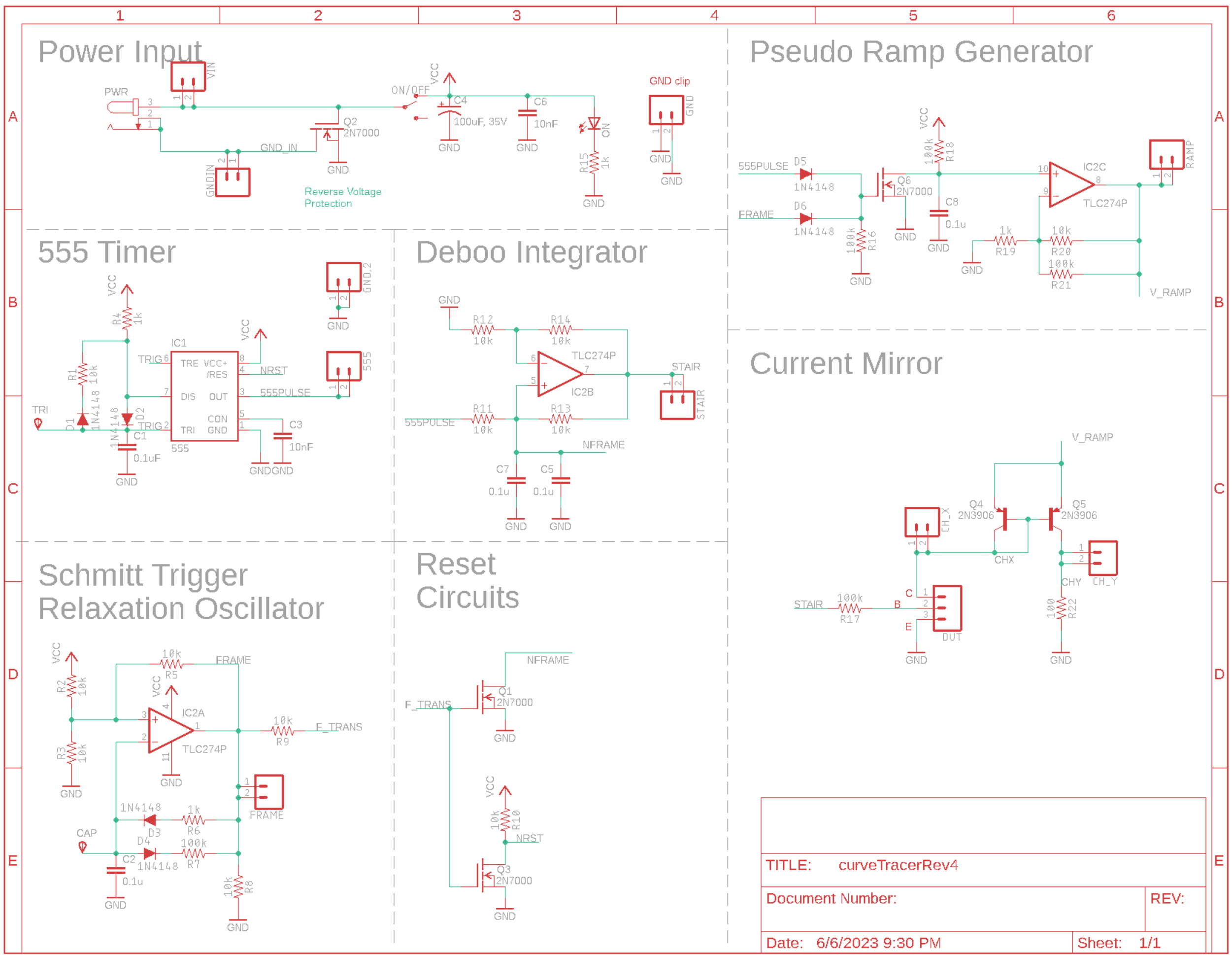
ELECTRICAL ENGINEERING DEPARTMENT
COLORADO SCHOOL OF MINES

EENG 385 Workbook

Created by Dr. Christopher Coulston

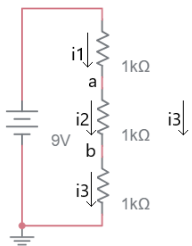
Spring 2026



Chapter: Review

Circuits with resistors, voltage and current sources

- Write your answers in the table shown on the next page. Write “unknown” in the entries where current/voltage cannot be determined (use when a value is infinite).
- On a separate page, show your work for each problem.
- Round you answer to 2-significant figures and include units.



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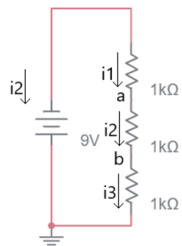

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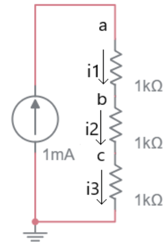

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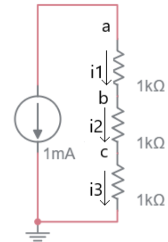

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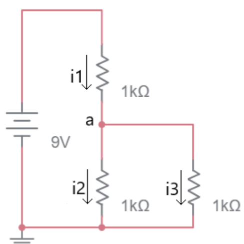

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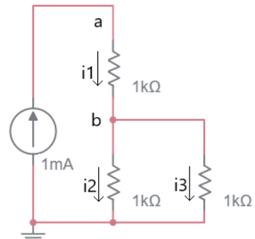

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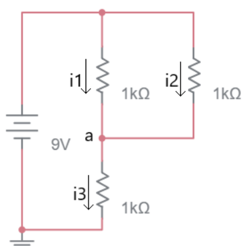

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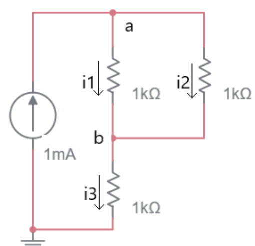

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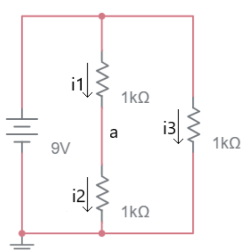

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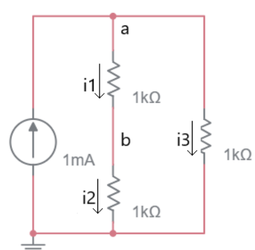

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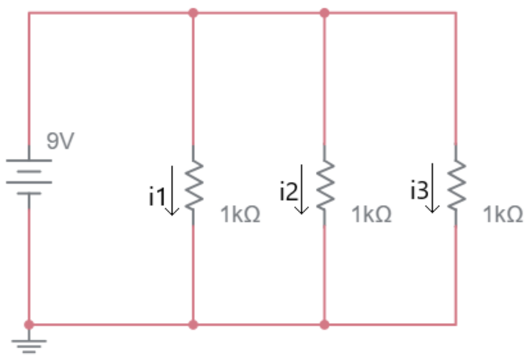

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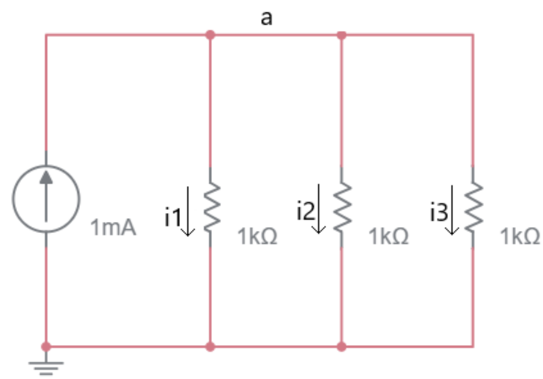

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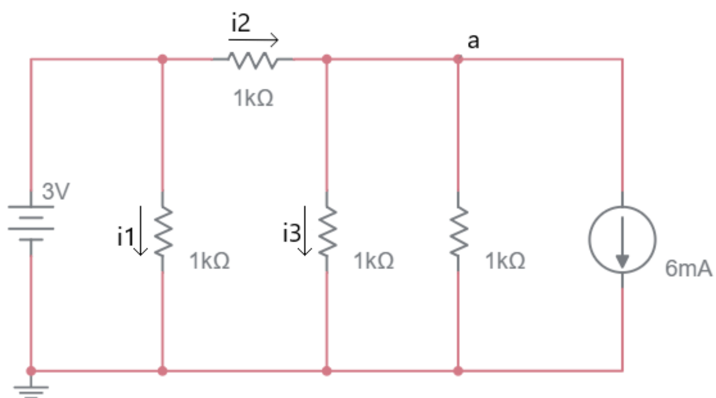

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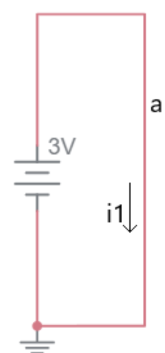

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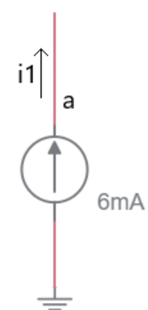


015

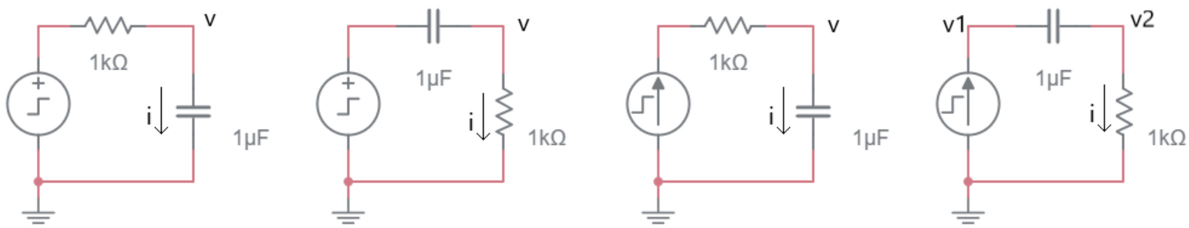
Circuit		a	b	c	i1	i2	i3
001							
002							
003							
004							
005							
006							
007							
008							
009							
010							
011							
012							
013							
014							
015							

Circuits with resistors and capacitors

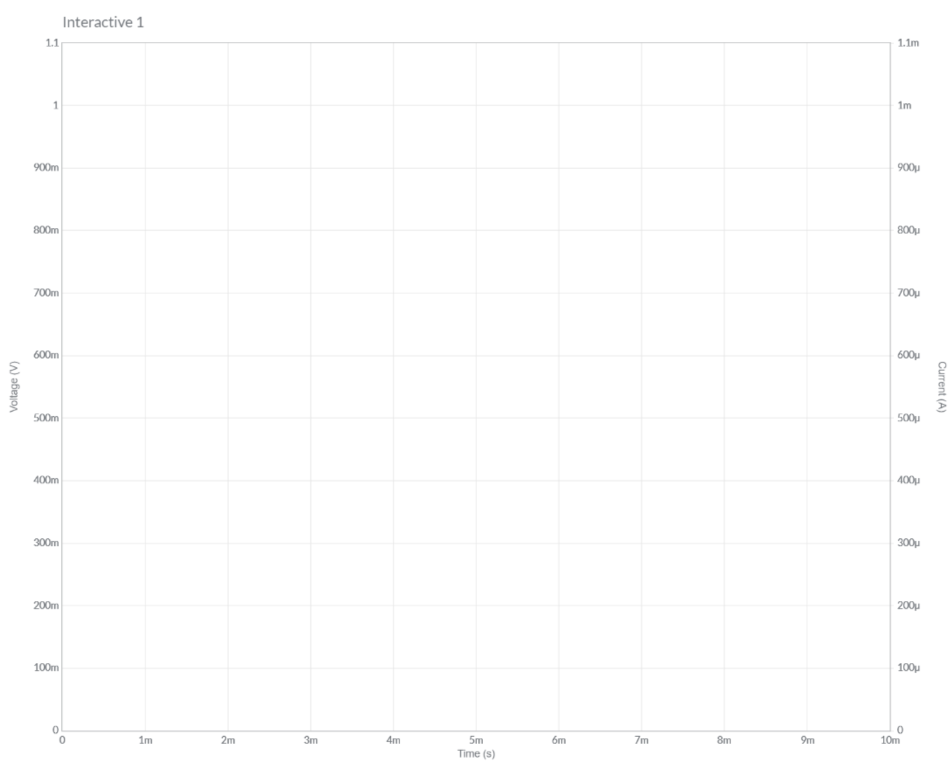
Plot all voltage across the capacitor and the current through the capacitor as a function of time. Assume that the capacitor is initially discharged; in other words, the voltage across the capacitor is 0V at time 0.

The  symbol is a unit step voltage source that goes from 0V to 1V at time 0.

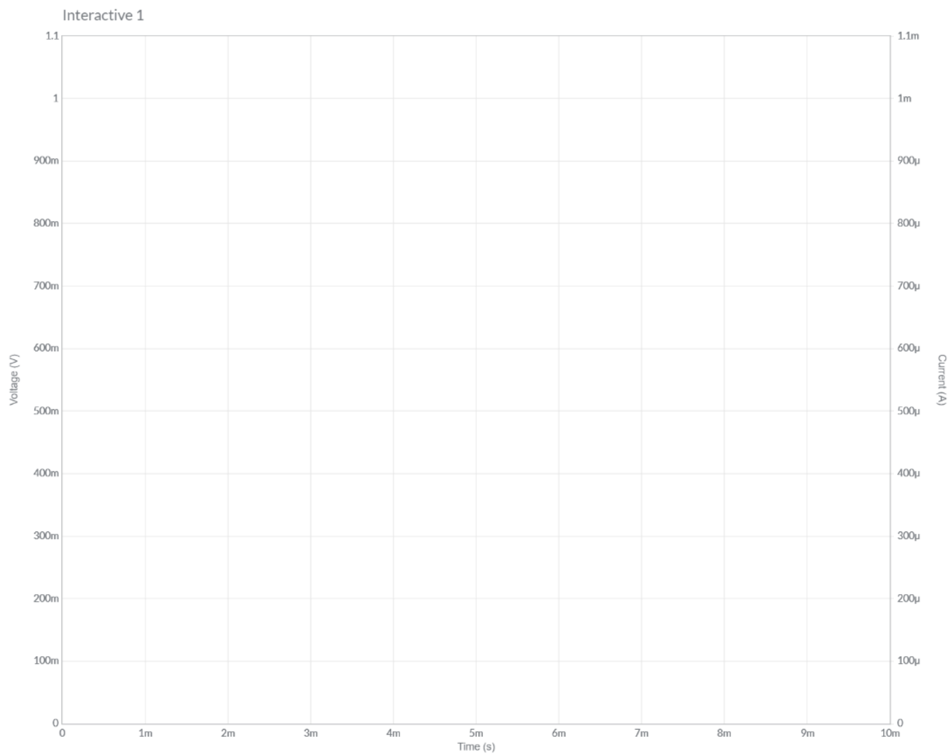
The  symbol is a unit step current source that goes from 0V to 1mA at time 0.



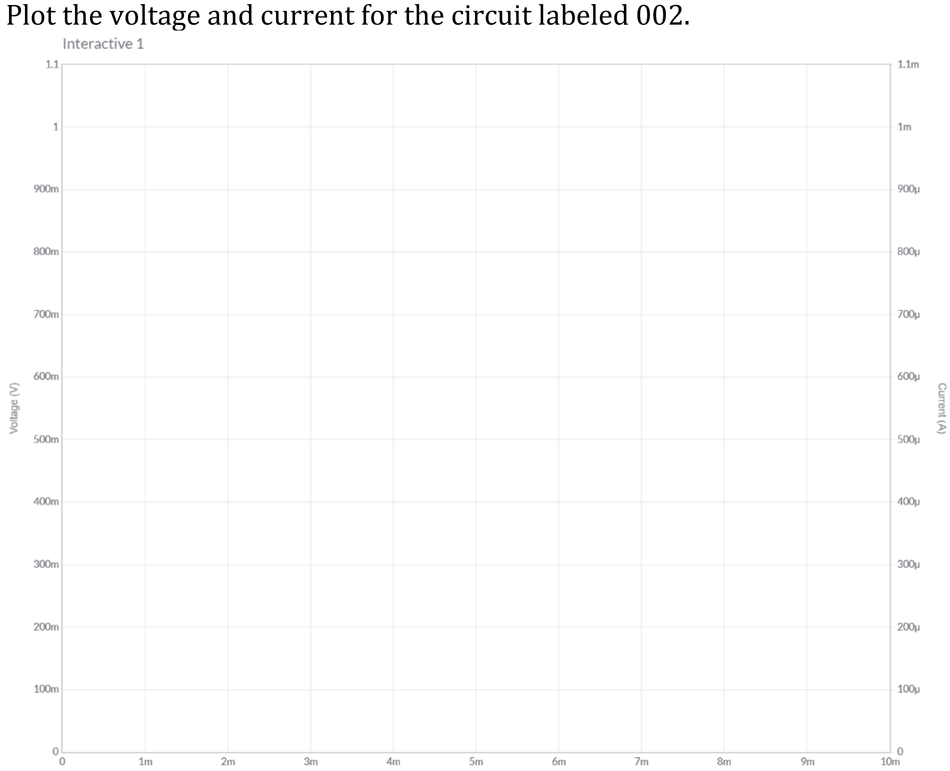
001 002 003 004



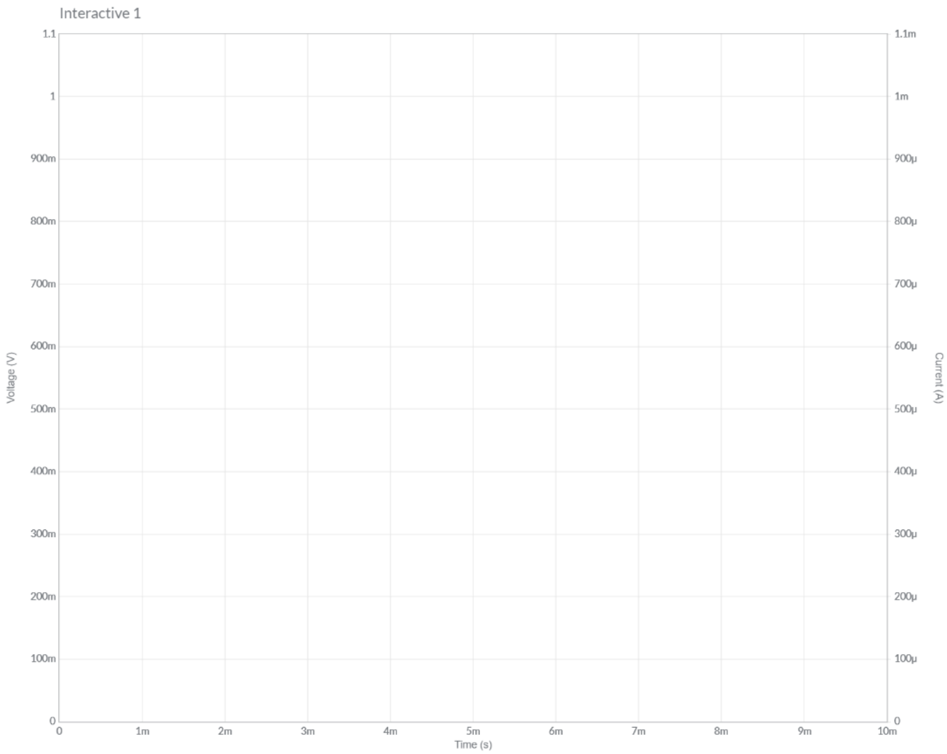
Plot the voltage and current for the circuit labeled 001.



Plot the voltage and current for the circuit labeled 002.

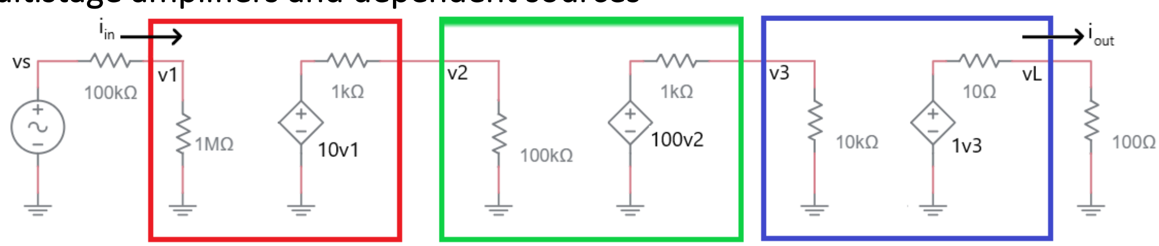


Plot the voltage and current for the circuit labeled 003.



Plot the voltage and current for the circuit labeled 004.

Multistage amplifiers and dependent sources



Determine the voltage gain v1/vs using the following steps

Step 0: Determine the gain v1/vs by examining the voltage divider formed by the source and the input to the first (red) stage.

Step 1: Determine the gain v2/v1 by examining the voltage divider formed by the output of the first stage (red) and the input to the second stage (green).

Step 2: Determine the gain v3/v2 by examining the voltage divider formed by the output of the second stage (green) and the input to the third stage (blue).

Step 3: Determine the gain vL/v3 by examining the voltage divider formed by the output of the third stage (blue) and the load resistor.

Step 4: Multiple the ratios from steps 0 to 3 together to fine the overall gain vL/vs.

Determine the current gain $A_i = i_{out}/i_{in}$

Step 1: Use Ohm's law at the input to write an equation for i_{in} in terms of v1 and the resistance.

Step 2: Use Ohm's law at the output to write an equation for i_{out} in terms of VL and the resistance.

Step 3: Use the previous two steps to write the ratio i_{out}/ i_{in} in terms of v1 and vL. These use the results from the previous section to arrive at a numerical value for the current gain.

Determine the power gain $A_p = p_{out}/p_{in}$

Use the previous two results to describe p = v*i for the output and input.

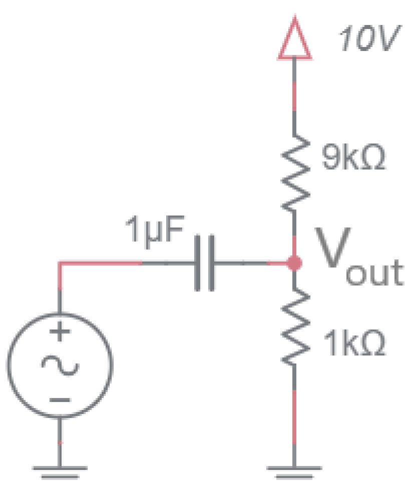
DC and AC simplifications for common circuit elements

As an engineer you will often need to make quick estimate of a circuit's behavior. In these cases, it is completely appropriate to make approximations of component behavior to simplify the analysis.

Element	DC Model	AC Model
Resistor	R	R
Capacitor	Open	C
Inductor	Short	L
Voltage Source	V	Short
Current Source	I	Open

Practical Application – Coupling AC signal on a DC bias

The following circuit is very useful as it allows an AC signal to be DC biased at any level you need. In order to understand how this circuit operates, you will need to invoke the principle of superposition. Superposition dictates that the total response of a system is the sum of the partial response. A partial response is the contributions that one energy source has on the output while all the other energy sources are set to 0. So, in terms of the schematic given the total response is the sum of the partial response due to the AC source (with the 10V supply set to 0V), plus the partial response due to the DC source (with the AC source set to 0).



In this analysis, assume that the AC source is described by 0.5sin(6,280t), a 1V peak to peak 1kHz sine wave.

In the Draw the schematic row, draw the equivalent circuit for each source. For example:

- In the AC Source column draw the schematic with all components replaced by their AC equivalent model from the table in the **DC and AC simplifications for common circuit elements** section and setting the 10V supply to 0V.
- In the DC Source column draw the schematic with all components replaced by their DC equivalent model from the table in the **DC and AC simplifications for common circuit elements** section and setting the AC source to 0V.

In the Vout row, use circuit theory to determine the value of Vout.
In the Total Response row add together the AC and DC contributions.

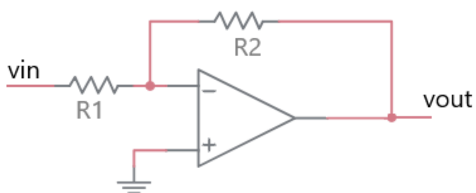
	AC Source (Set DC source to 0)	DC Source (Set AC source to 0)
Draw the schematic		
Vout		
Total Response		

Chapter: Opamp

Circuits with opamps and resistors

Determine the gain (vout/vin) for circuits 001 and 003. Compute vout in terms of v1, v2, v3 for circuits 002 and 004.

- Write your answers in the table shown.
- On a separate page, show your work for each problem.
- Write "unknown" for circuits whose current/voltage cannot be determined.



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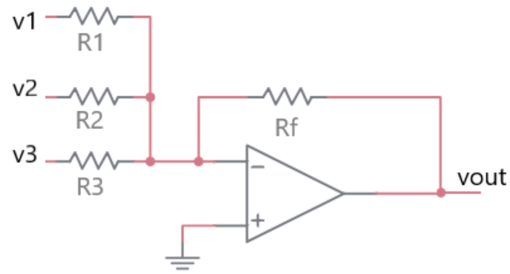

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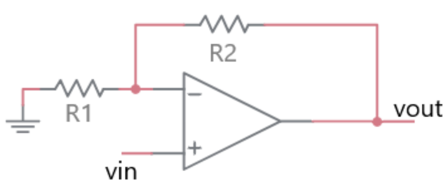
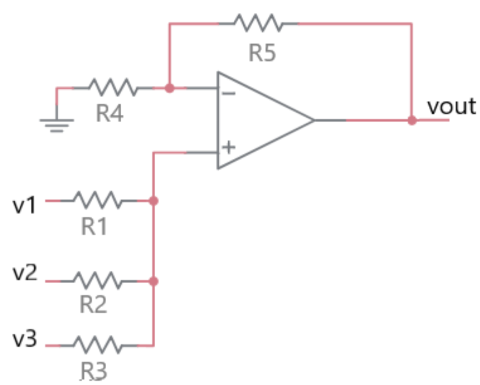


003



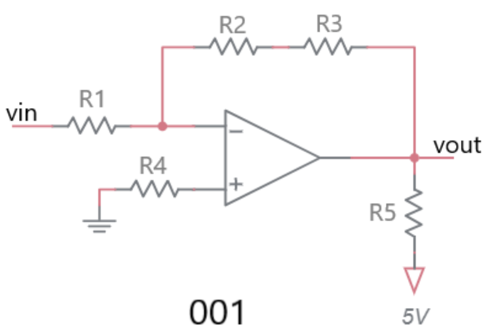
004

Circuit	Gain or vout
001	
002	
003	
004	

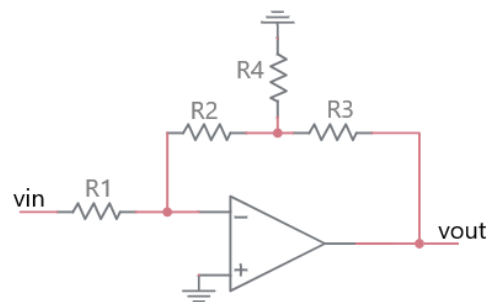
Circuits with opamps and resistors

Determine the gain (vout/vin) for the following circuits.

- Write your answers in the table shown.
- On a separate page, show your work for each problem.
- Write "unknown" for circuits whose current/voltage cannot be determined.



001

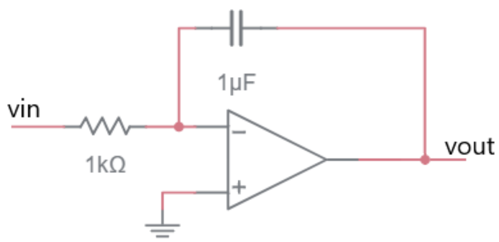


002

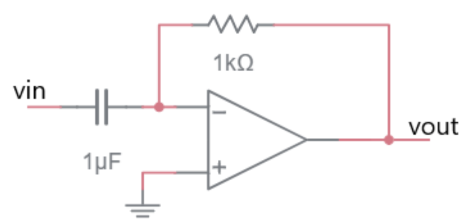
Circuit	Gain
001	
002	

Circuits with opamps, capacitors and resistors

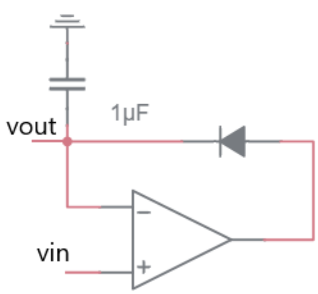
For each of the circuits, plot vout in terms of the input signal given.



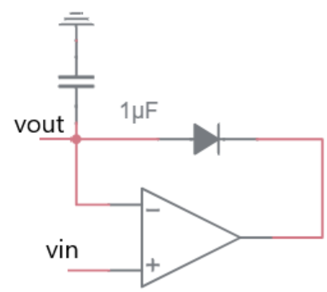
001



002

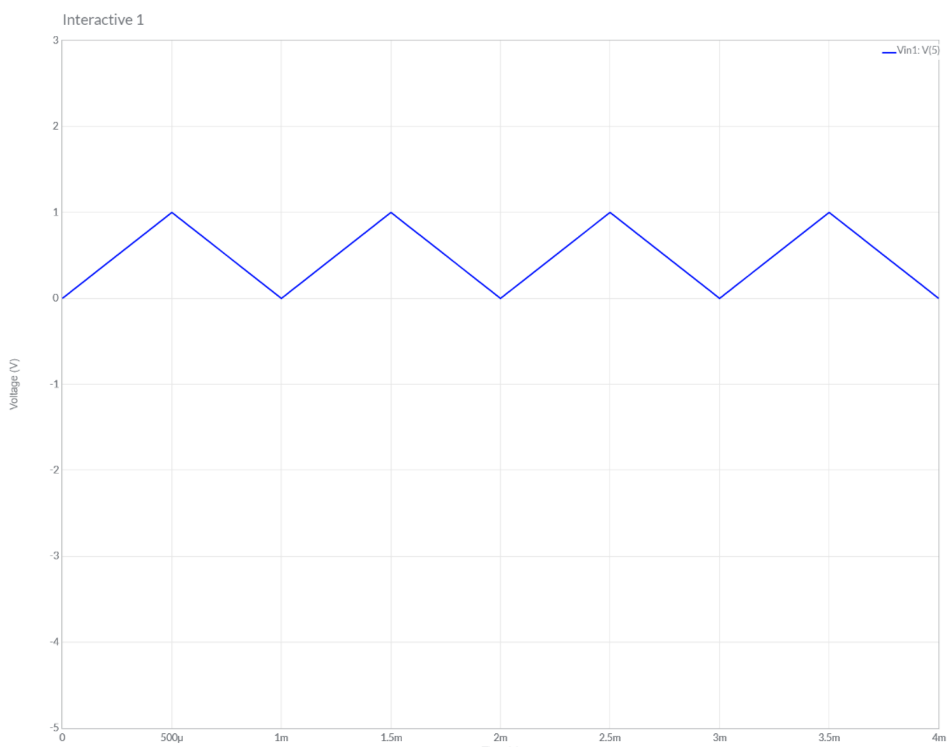


003

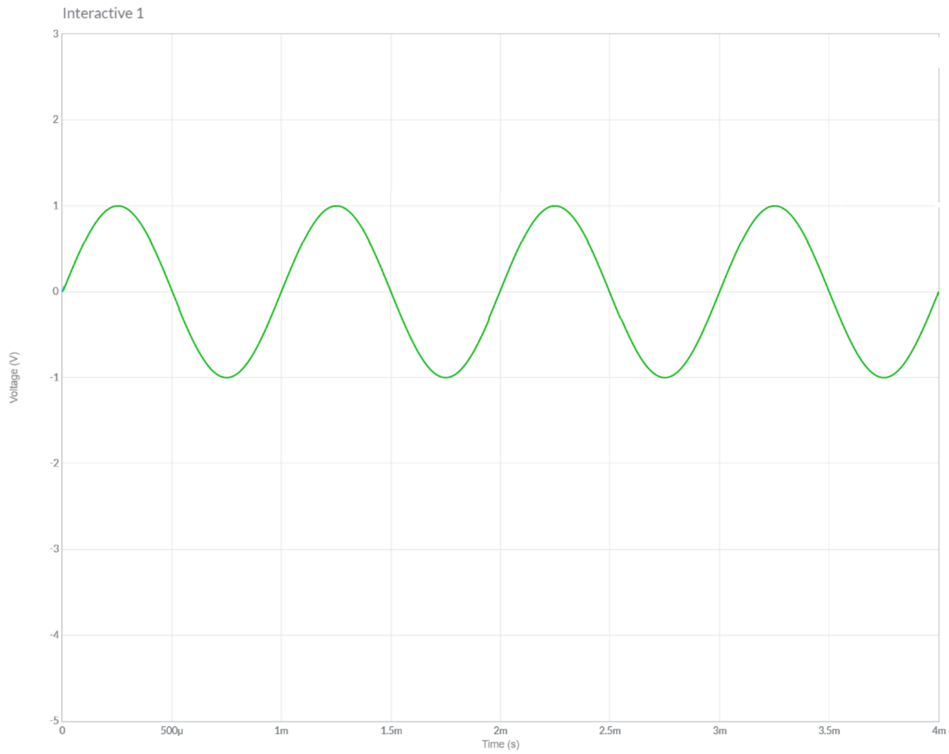


004

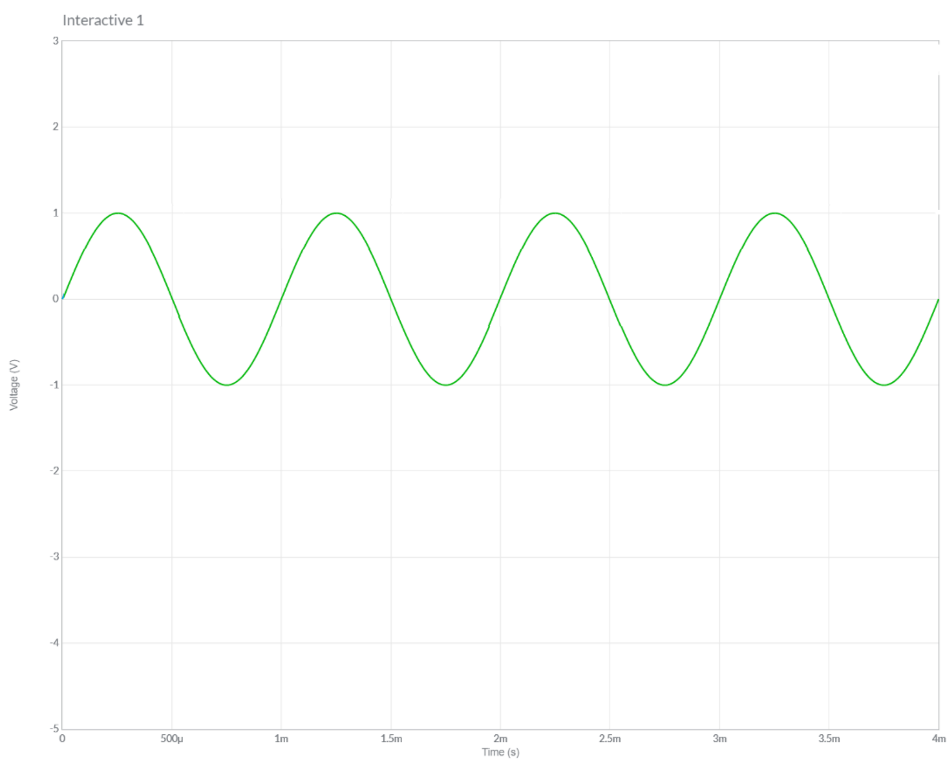
Plot vout for the circuit labeled 001.



Plot vout for the circuit labeled 002.

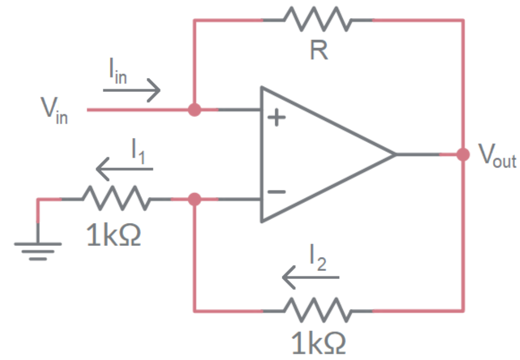


Plot vout for the circuit labeled 003.



Plot vout for the circuit labeled 004.

Practical Application - Negative Impedance Converter



The negative impedance converter (NIC) is an active circuit which injects energy into circuits in contrast to an ordinary load that consumes energy from them. Let's analyze the NIC shown to better understand what this means. Fill in your answers in the boxes provided.

1. Assume that the negative feedback makes the inverting and non-inverting inputs of the opamp equal. What is the voltage on the non-invering input of the opamp?

2. Use Ohm's Law to determine the current I_1.

3. Use KCL to determine the current I_2.

4. Use Ohm's Law to determine V_{out}. Express your answer in terms of V_{in}.

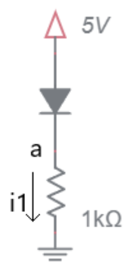
5. Write the Ohm's Law expression for the resistor R. Take care with the passive sign convention and substitute Vout with the expression you got in step 4.

6. Solve the equation in step 5 for V_{in}/I_{in}.

Chapter: Diode

DC circuits with diodes and resistors

- Assume a constant voltage drop model for the diodes with a cut-in voltage of 0.7V.
- Write your answers in the table shown on the next page.
- On a separate page, show your work for each problem. Denote the bias of all diodes by writing FB/RB next to forward/reversed biased diodes respectively.
- Write “unknown” for circuits whose current/voltage cannot be determined.



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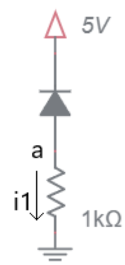

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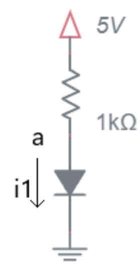

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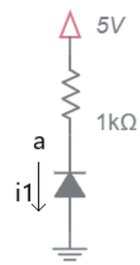

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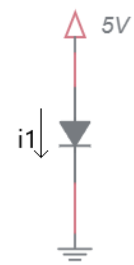

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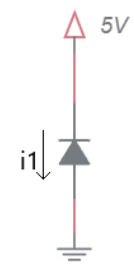

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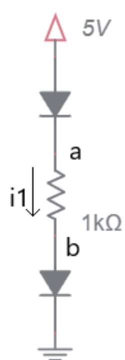

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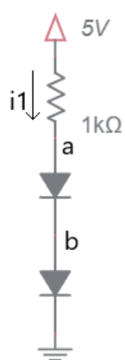

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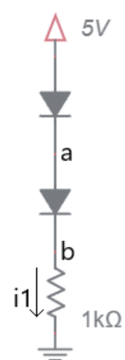

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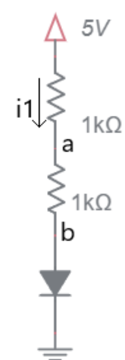

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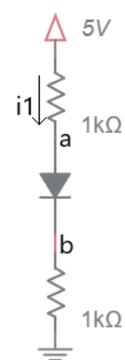

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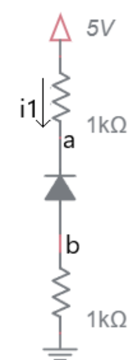

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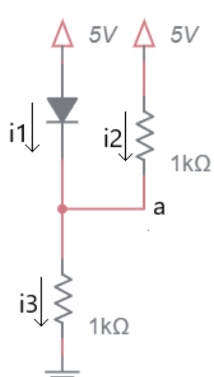

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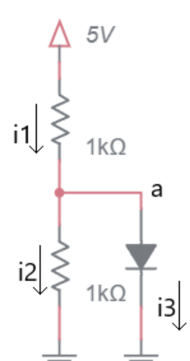

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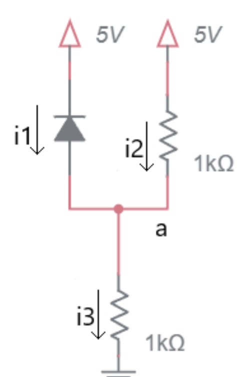

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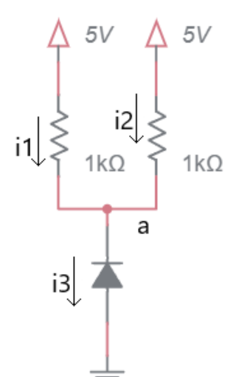


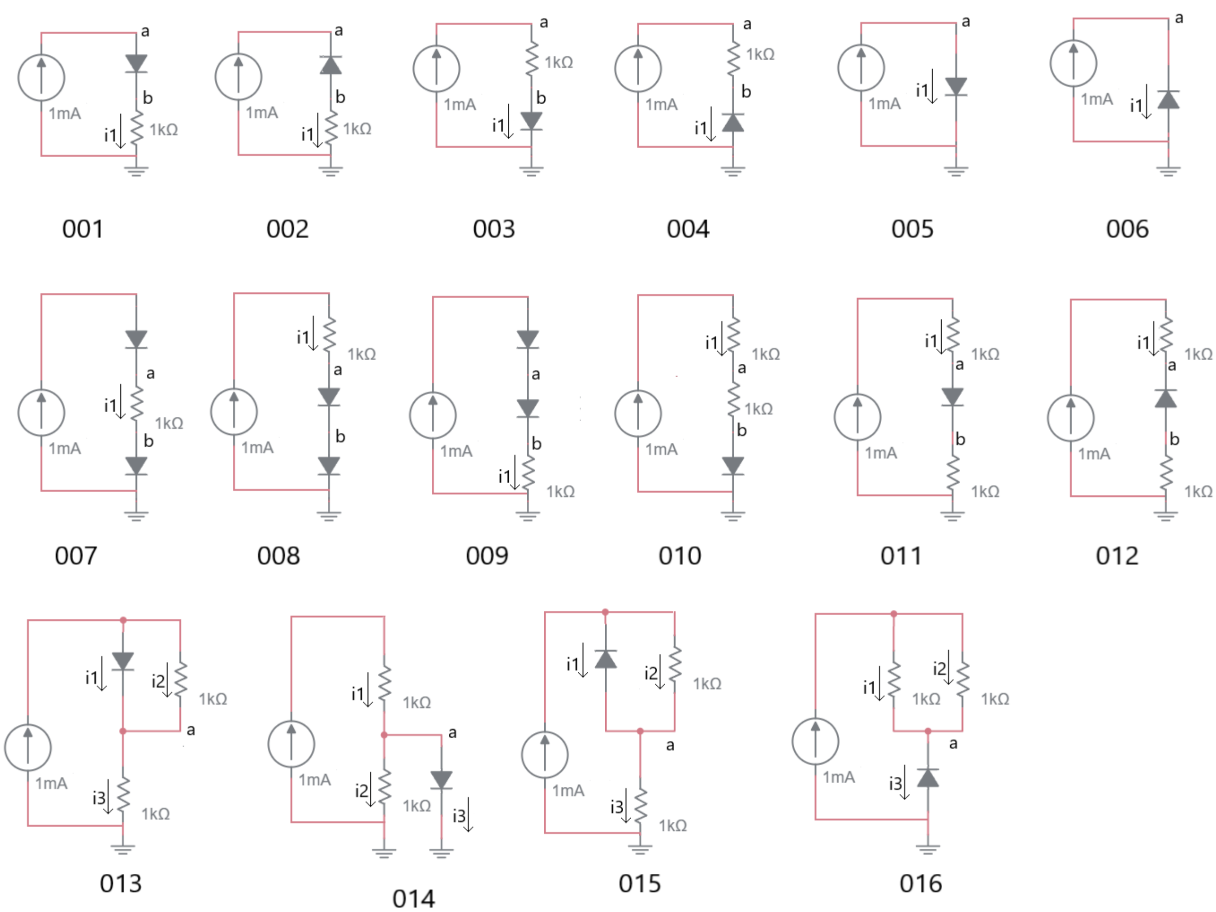
016

Problem		a	b	i1	i2	i3
001						
002						
003						
004						
005						
006						
007						
008						
009						
010						
011						
012						
013						
014						
015						
016						

Current Source circuits with diodes and resistors

Determine the currents and voltages in the following circuit.

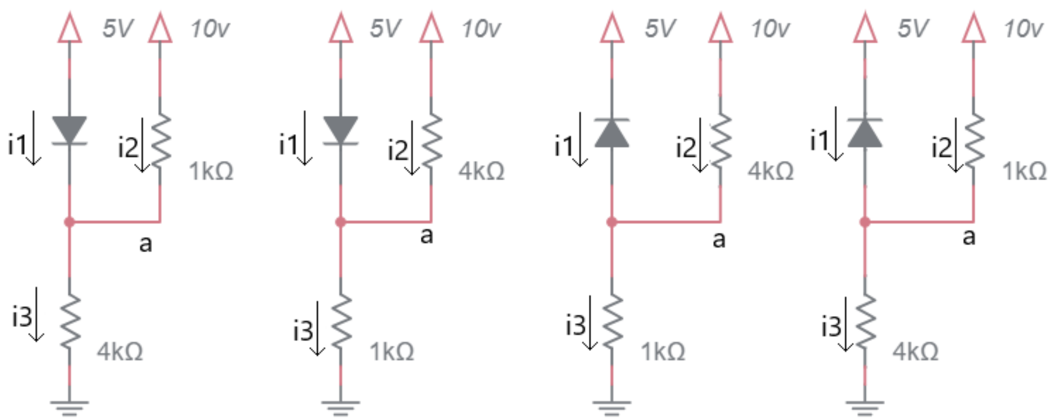
- Assume a constant voltage drop model for the diodes with a cut-in voltage of 0.7V.
- Write your answers in the table shown on the next page.
- On a separate page, show your work for each problem. Denote the bias of all diodes by writing FB/RB next to forward/reversed biased diodes respectively.
- Write “unknown” for circuits whose current/voltage cannot be determined.
- Note, the current source energizing all the circuits!



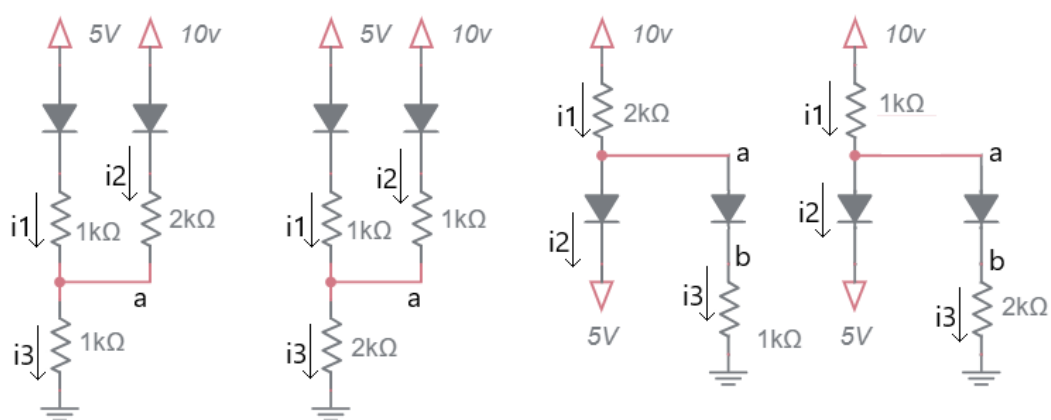
Problem		a	b	i1	i2	i3
001						
002						
003						
004						
005						
006						
007						
008						
009						
010						
011						
012						
013						
014						
015						
016						

DC circuits with diodes and resistors

- Assume a constant voltage drop model for the diodes with a cut-in voltage of 0.7V.
- Write your answers in the table shown on the next page.
- On a separate page, show your work for each problem. Denote the bias of all diodes by writing FB/RB next to forward/reversed biased diodes respectively.
- Write “unknown” for circuits whose current/voltage cannot be determined.
- Note, the power rails are a mix of 5V and 10V!



001 002 003 004

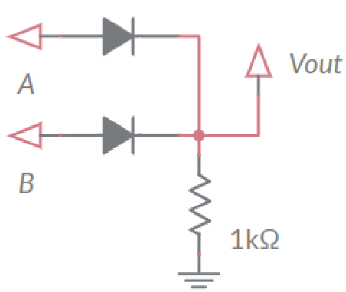


005 006 007 008

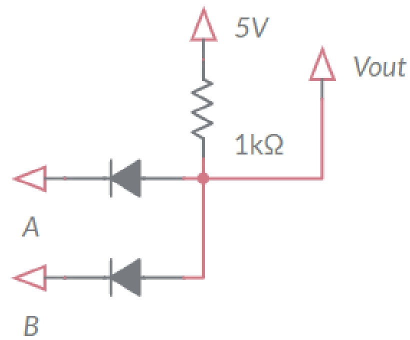
Problem		a	b	i1	i2	i3
001						
002						
003						
004						
005						
006						
007						
008						

Diodes as logic gates

In the following circuits consider 5V logic 1 and GND as logic 0. Use the ideal diode model for analysis. Complete the truth table for the circuit shown and describe the logic function instantiated by each.



001



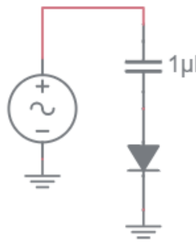
002

A	**B**	**Vout**
0	**0**	**0**
0	**1**	**1**
1	**0**	**1**
1	**1**	**1**

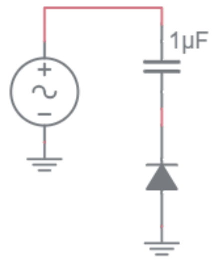
A	**B**	**Vout**
0	**0**	**0**
0	**1**	**0**
1	**0**	**0**
1	**1**	**1**

AC circuits with diodes and resistors

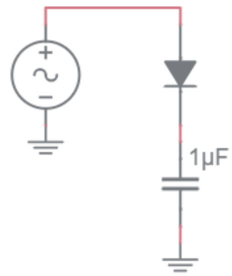
For each of the circuits below, complete the timing diagrams.



001



002



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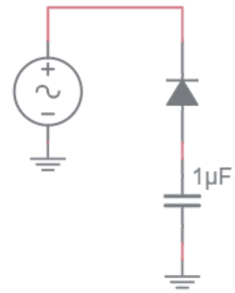
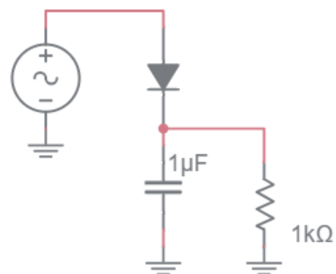
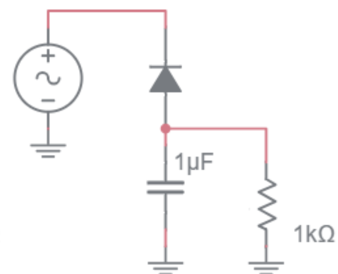


004



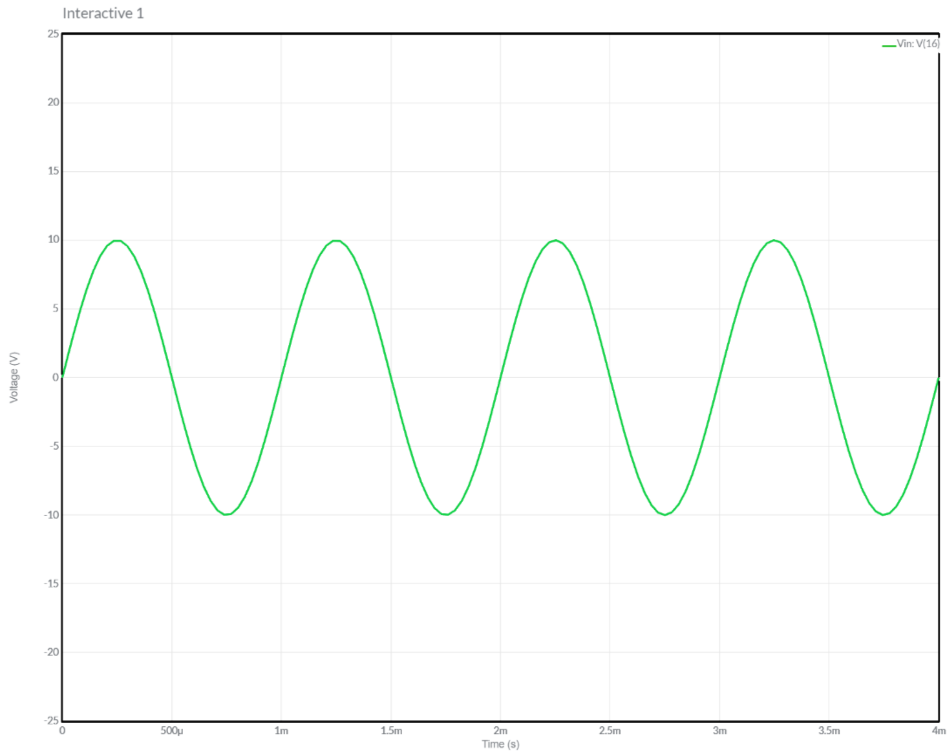
005



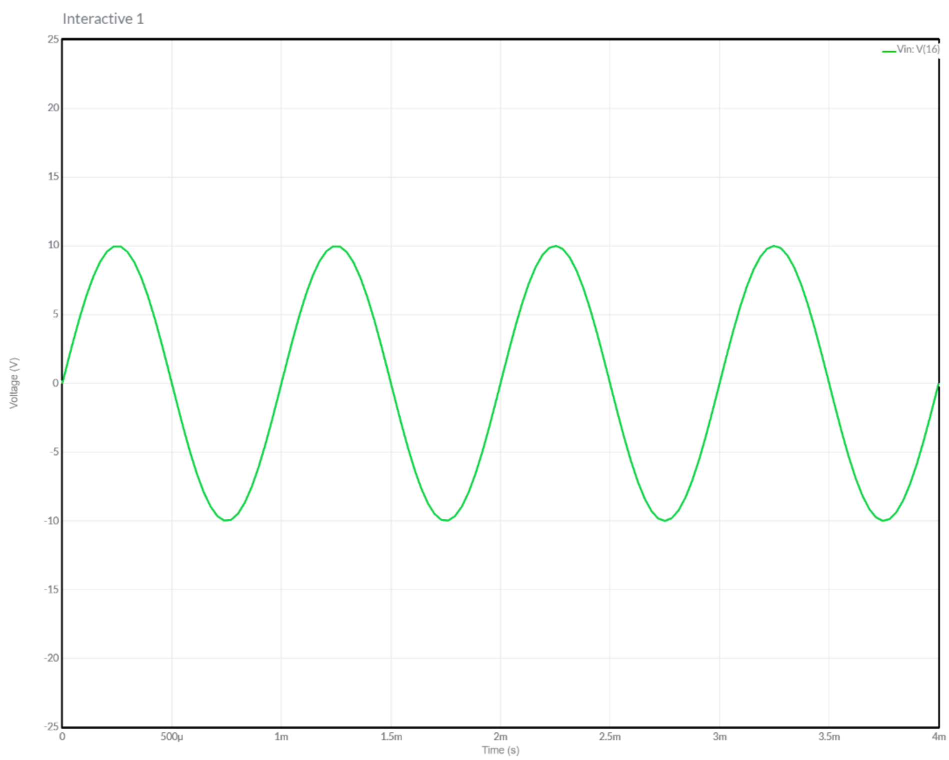
006

After you complete the timing diagrams for circuits 001 to 004, look at their behavior and assign the circuits one of the following names – one name for each circuit.

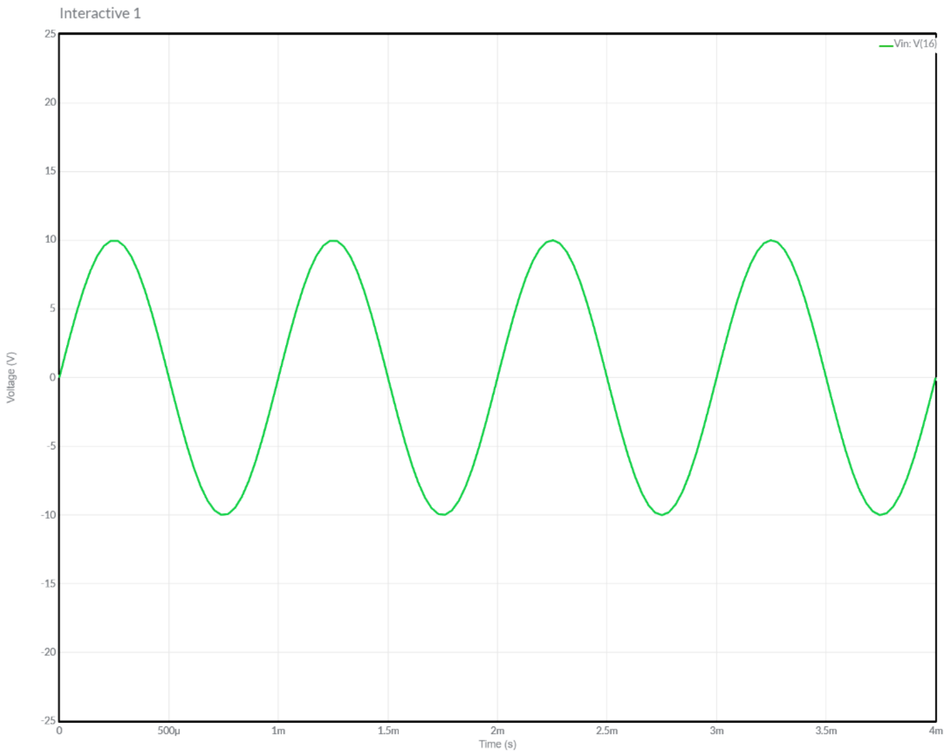
- Peak detector ________
- Positive level shifter ________
- Trough detector ________
- Negative level shifter ________



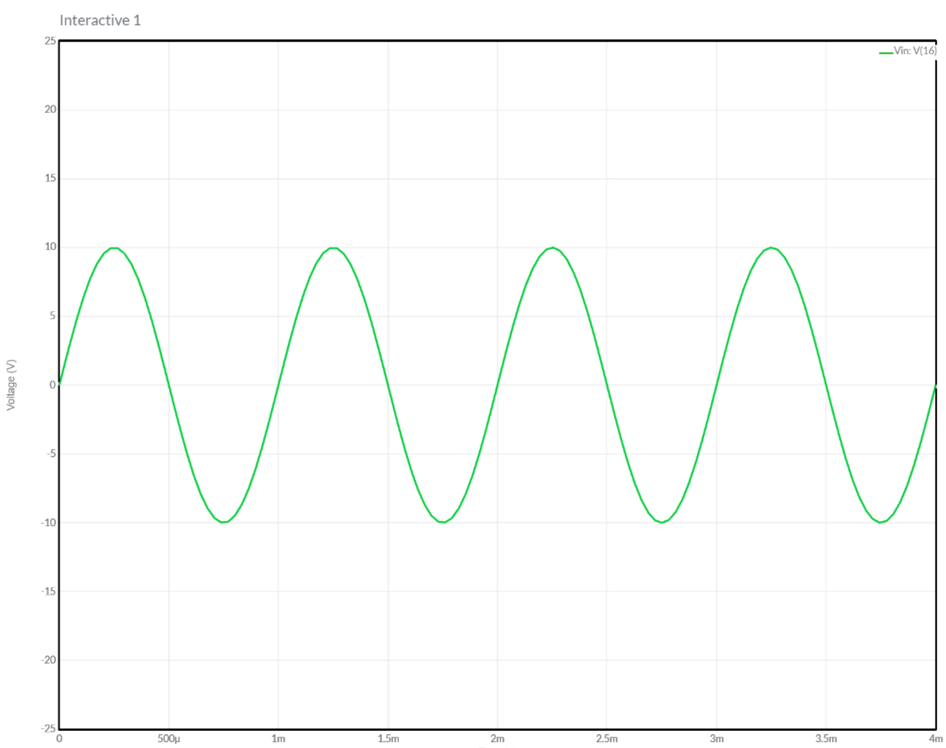
Plot the voltage for the circuit labeled 001.



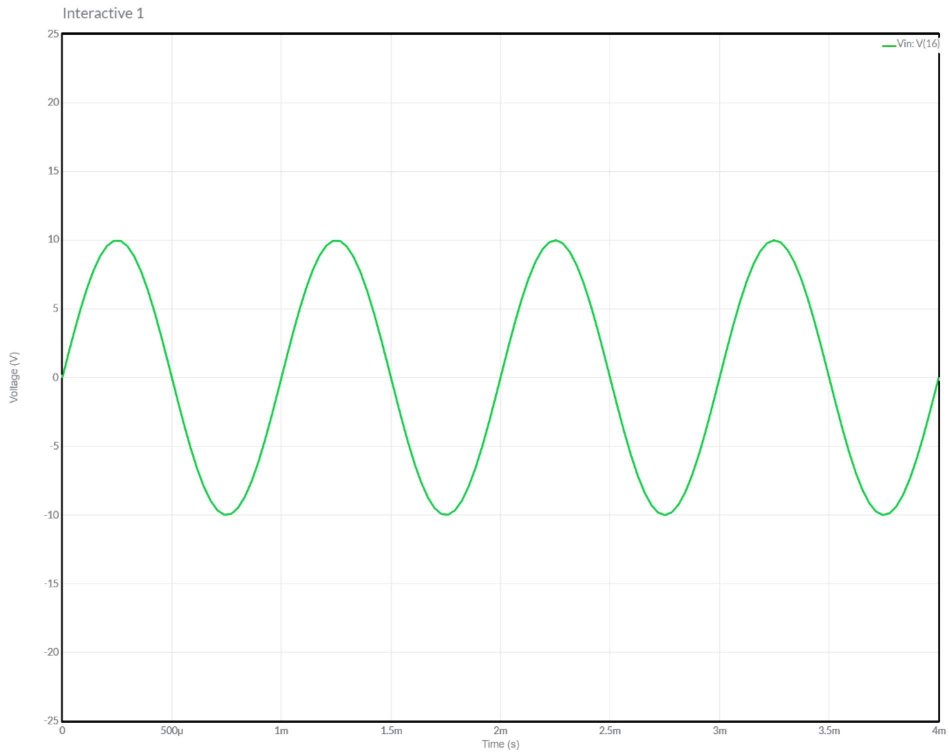
Plot the voltage for the circuit labeled 002.



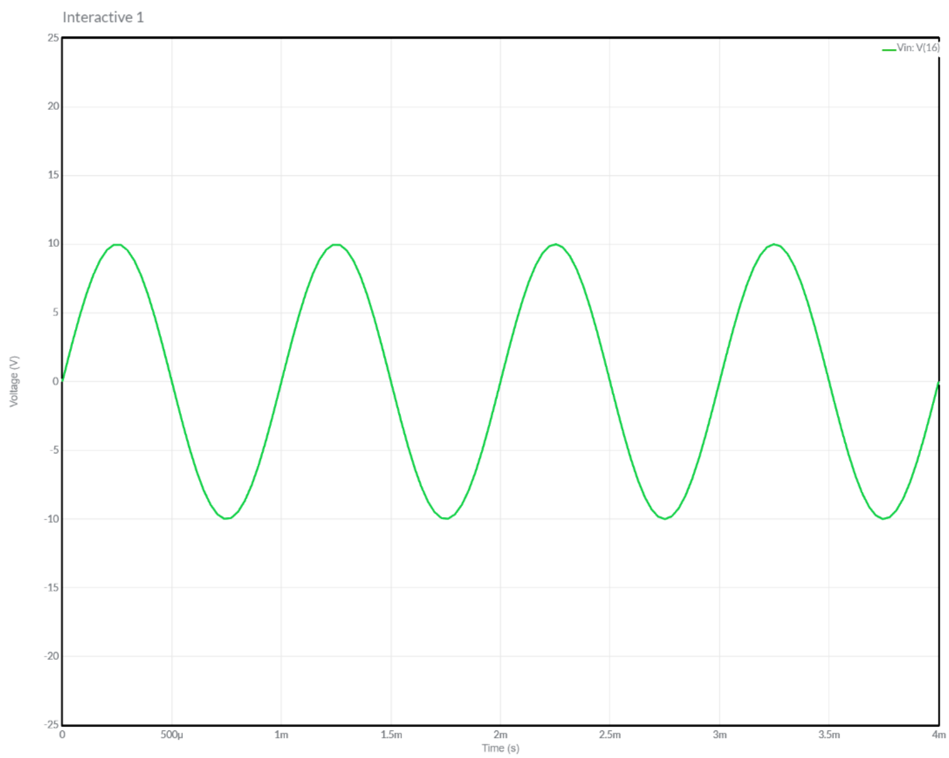
Plot the voltage for the circuit labeled 003.



Plot the voltage for the circuit labeled 004.



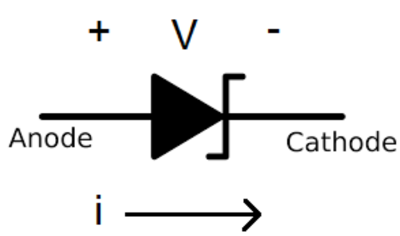
Plot the voltage for the circuit labeled 005.

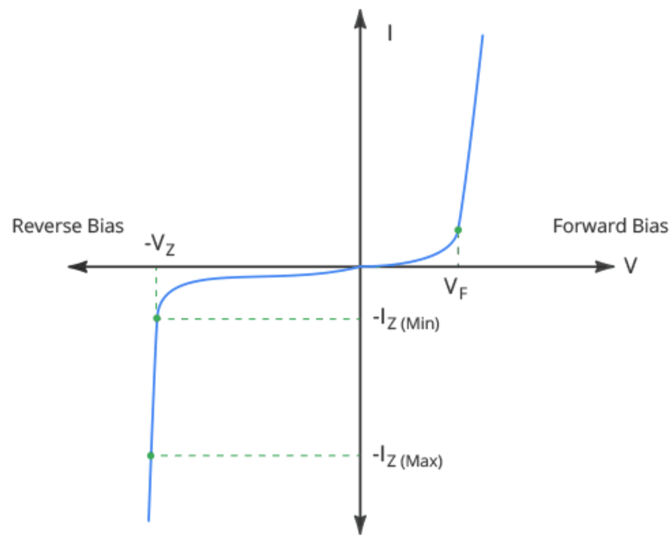


Plot the voltage for the circuit labeled 006.

Special Types of Diodes – The Zener Diode

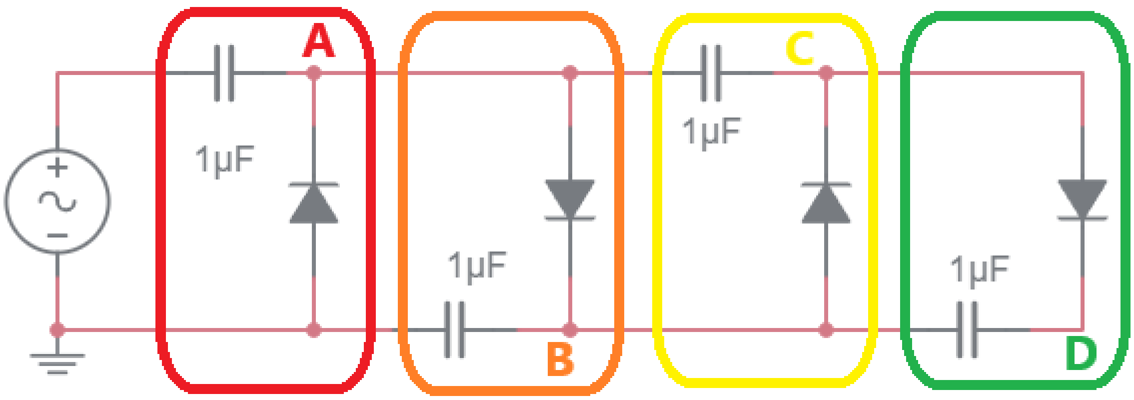
The Zener diode shown below is first and foremost a diode. In the I/V graph below, the forward biased Zener diode starts conducting current at V_F, typically about 0.7V. However, Zener diodes have a trick, when the reverse biased voltage exceeds the Zener diodes' Zener voltage it conducts current from the cathode to the diode while clamping the voltage at the Zener voltage $-V_Z$. Manufactures of Zener diodes produce them with a variety of Zener voltages. In order to maintain a clamped reverse biased voltage, a Zener diode must pass at least $-I_Z$ amps, the minimum Zener current.





Practical Application – Voltage Doubler

The circuit below takes combinations of circuits from the **AC circuits with diodes and resistors** section and combines them to produce a larger voltage than the input voltage. Before starting on this analysis, you must first have successfully completed all the problems in the **AC circuits with diodes and resistors** section.



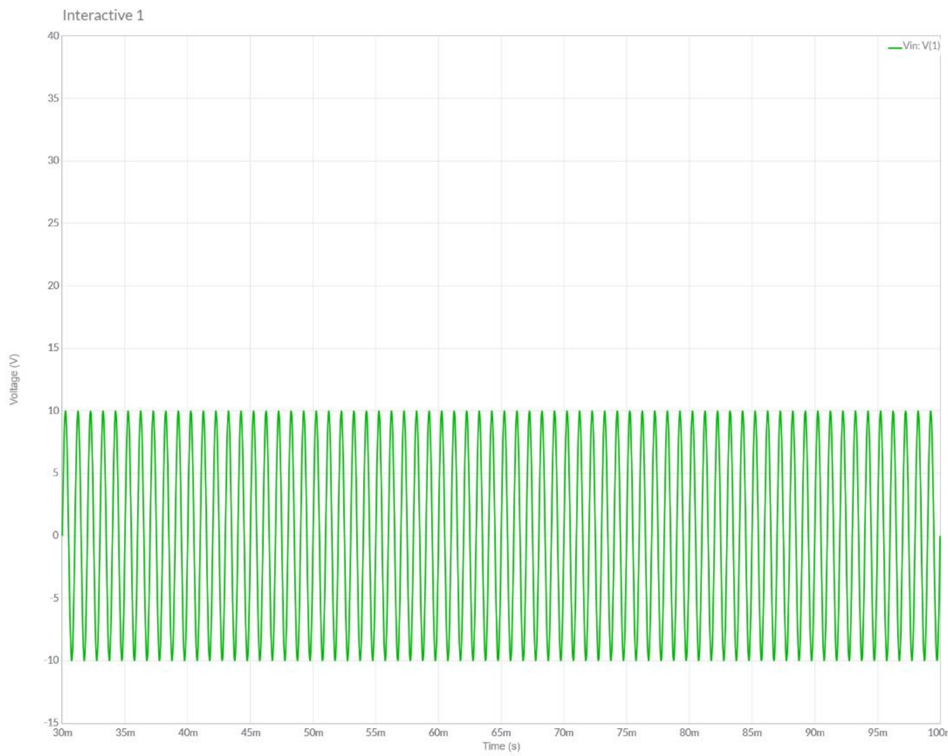
1. Look at the circuit outlined in red and compare it to circuits 001 to 004 in the **AC circuits with diodes and resistors** section. Which circuit is it and what name did you assign it?

2. The circuit outlined in orange gets it input from A. Look at the circuit outlined in orange and compare it to circuits 001 to 004 in the **AC circuits with diodes and resistors** section. Which circuit is it and what name did you assign it?

3. The circuit outlined in yellow gets it input from A and its ground terminal is connected to B. Look at the circuit outlined in yellow and compare it to circuits 001 to 004 in the **AC circuits with diodes and resistors** section. Which circuit is it and what name did you assign it?

4. The circuit outlined in green gets it input from C and its ground terminal is connected to B. Look at the circuit outlined in green and compare it to circuits 001 to 004 in the **AC circuits with diodes and resistors** section. Which circuit is it and what name did you assign it?

Use this information to plot the output of each colored stage given the input shown.


Interactive 1
Vin: V(1)
Voltage (V)
Time (s)
40
35
30
25
20
15
10
5
0
-5
-10
-15
30m
35m
40m
45m
50m
55m
60m
65m
70m
75m
80m
85m
90m
95m
100m

Practical Application – I/V Curve for LED

You are required to illuminate a Cree Xlamp XM-L2 High Power LED using either a current source or voltage source, the choice is yours. The LED has a I vs. V curve is shown in Figure 1. You are tasked to operate the LED at 3V and 1.3A – the point in the middle of the curve.

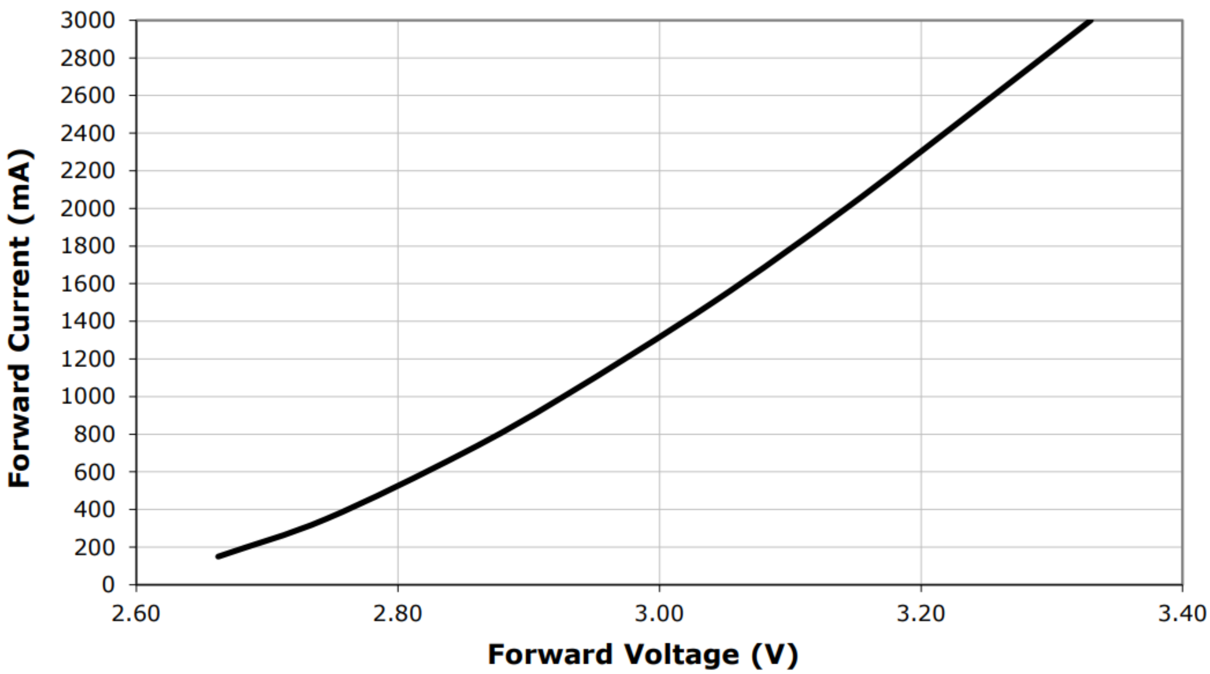


Figure 1: I/V curve for a Cree Xlamp XM-L2 High Power LED.

Compute the variation in power using a voltage source with a ±10% variation in voltage.

a. Using this voltage source, what is the range of voltages across the LED?

b. Using the I/V curve in Figure 1, what is the range of current through the LED?

c. What is the range of power dissipated by the LED voltage source?

Compute the variation in power using a current source, with a ±10% variation in current.

a. Using this supply, what is the range of current through the LED?

b. Using the I/V curve in Figure 1, what is the range of voltage across the LED?

c. What is the range of power dissipated by the LED using the current source?

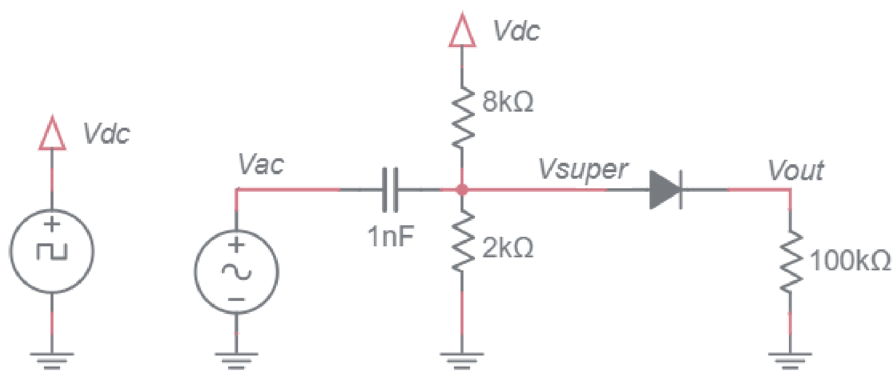
@1.17A, 2.95V	**@1.43A, 3.05V**

What supply (voltage and current) produces the least variation in power?

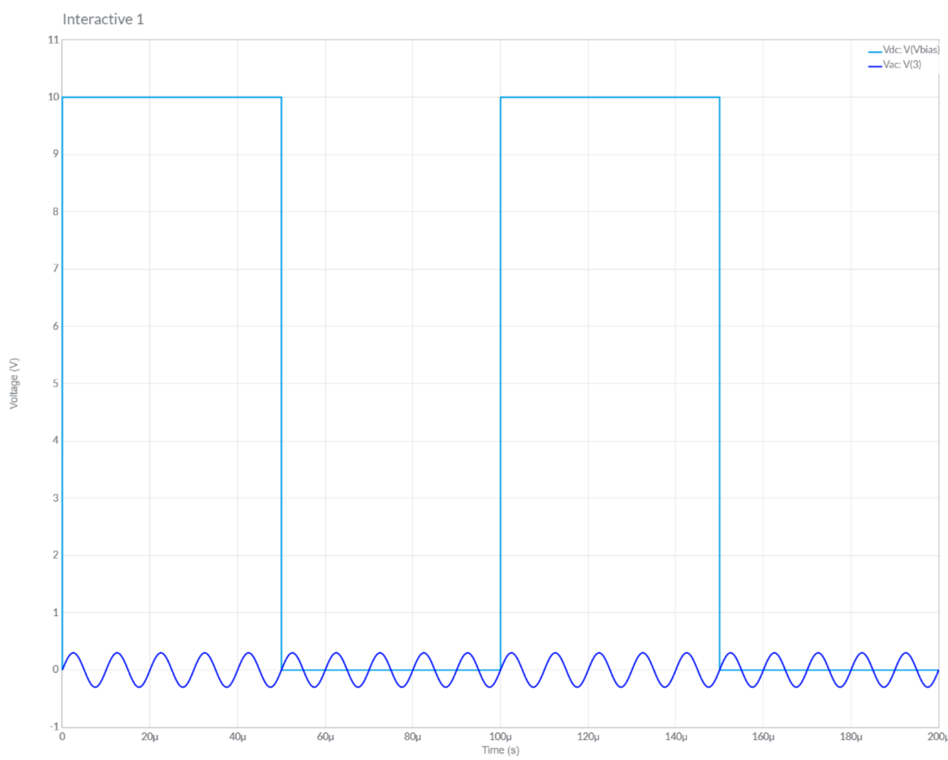
What characteristic of the I/V curve determines the sensitivity to changes in voltage or current?

Practical Application – Using diodes as a switch

The following circuit shows how you can use a diode to switch a (small) AC signal on or off.



Your first task is to determine the signal *Vsuper* from the *Vac* and *Vdc* signals. To do this ignore the diode and the 100kΩ resistor and use superposition to determine Vout. Note that the *Vdc* signal is a square wave that alternates between 0V and 10V at 10kHz (see graph below). If you are having a problem with this open the workbook to the Review section and find the **Practical Application – Coupling AC signal on a DC bias** section. You could also use the simulator for some hints.



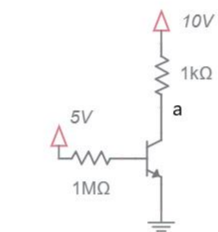
Note compute the *Vout* by running the *Vsuper* signal through a diode modeled as a constant voltage drop. You should see that when the *Vdc* signal is at 10V the AC signal passes through to *Vout*. When the *Vdc* signal is at 0V, *Vout* = 0V.

Chapter BJT @ DC

DC circuits with NPN BJTs and resistors

Determine the voltage at the labeled nodes. Determine collector, base and emitter currents.

- If the BJT is in the active region, assume V_{be} = 0.7V, ß = 100, and i_c=i_e.
- If the BJT is in saturation, assume that V_{ce} = 0.2V.
- Only compute ß when the BJT is in saturation.
- Write your answers on the table shown on the page following the problems.
- On separate page(s), show your work for each problem. Denote the operating region of BJTs by writing ACT/SAT next to active/saturated mode BJTs respectively.
- Write "unknown" for circuits whose current/voltage cannot be determined.
- Round you answer to 2-significant figures and include units.



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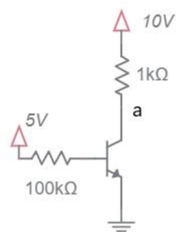

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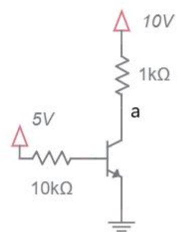

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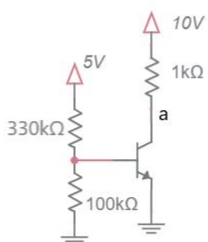

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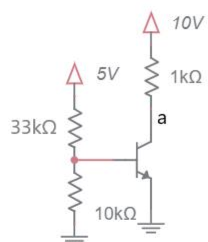

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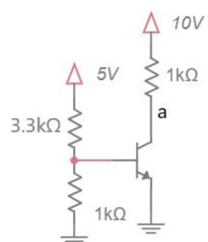

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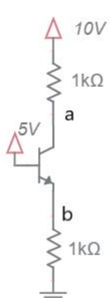

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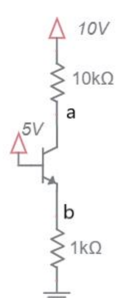

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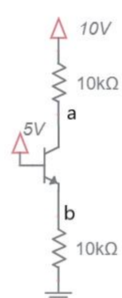

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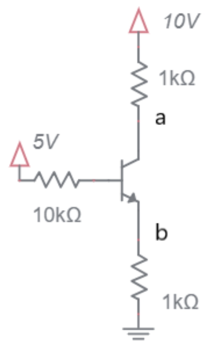

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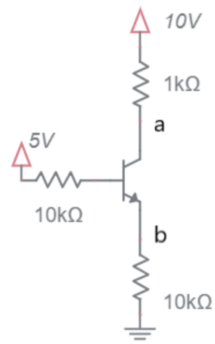

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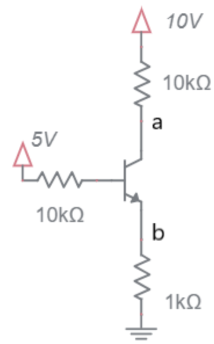

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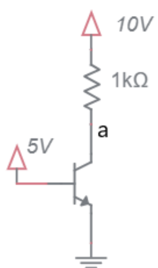

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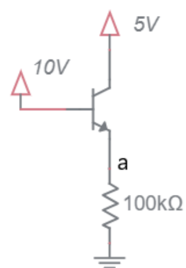

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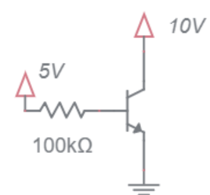

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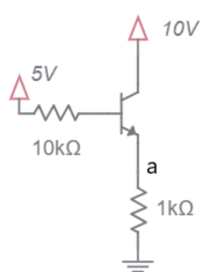

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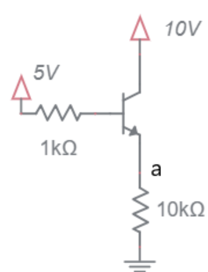


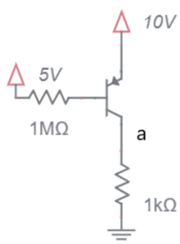
017

Problem		a	b	ic	ib	ie	ß
001							
002							
003							
004							
005							
006							
007							
008							
009							
010							
011							
012							
013							
014							
015							
016							
017							

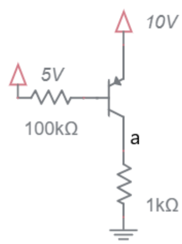
DC circuits with PNP BJTs and resistors

Determine the voltage at the labeled nodes. Determine collector, base and emitter currents.

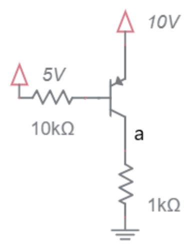
- If the BJT is in the active region, assume V_{eb} = 0.7V, ß = 100, and i_c=i_e.
- If the BJT is in saturation, assume that V_{ec} = 0.2V.
- Only compute ß when the BJT is in saturation.
- Write your answers on the table shown on the page following the problems.
- On separate page(s), show your work for each problem. Denote the operating region of BJTs by writing ACT/SAT next to active/saturated mode BJTs respectively.
- Write "unknown" for circuits whose current/voltage cannot be determined.
- Round you answer to 2-significant figures and include units.



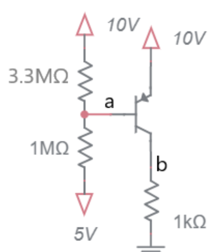
001



002



003



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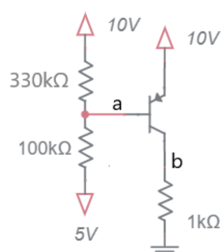
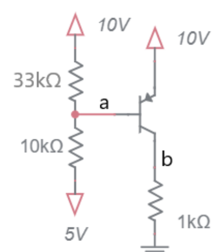
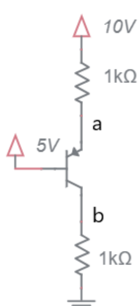


005



006



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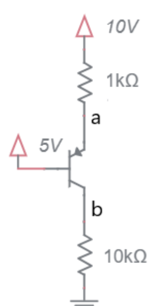

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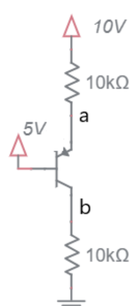
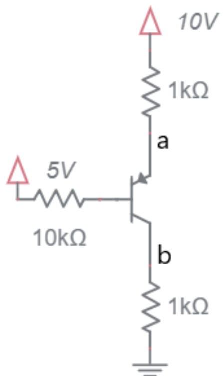
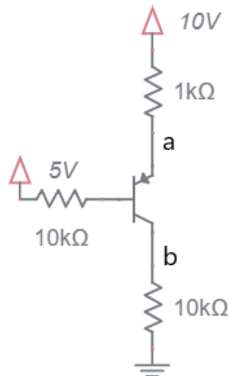


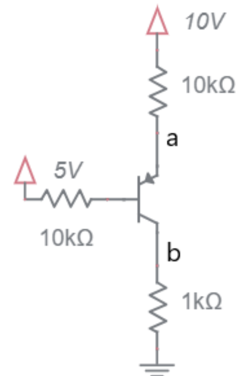
009



010



011



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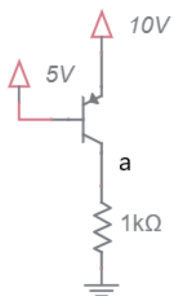

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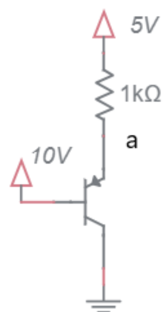
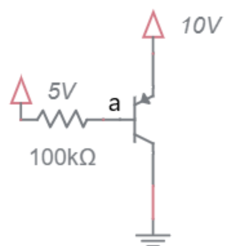


014

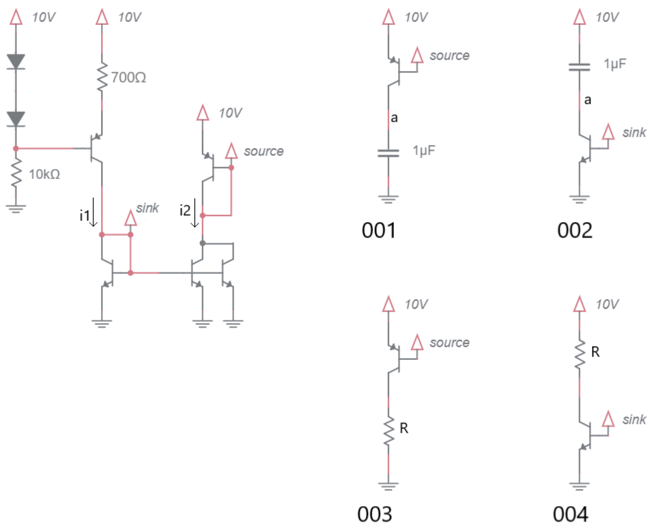


015

Problem		a	b	ic	ib	ie	ß
001							
002							
003							
004							
005							
006							
007							
008							
009							
010							
011							
012							
013							
014							
015							

BJT Current Sources and mirrors

If the BJT is in the active region, assume V_{be} = 0.7V, ß = 100, i_c=i_e.



1. Determine the currents i1 and i2.

2. What is the range (upper and lower) of resistance in the circuits 003 and 004 to keep their BJTs in the active region?

3. Plot the volage at point **a** in circuits 001 and 002 assuming the capacitor is discharge at time 0. Use the following questions to guide yourself to the answer.
   a. Using the voltage/current equation for a capacitor, find dv/dt for 001 and 002.

   b. Since the initial voltage drop across the capacitors is 0V, what is the initial value of **a**?

c. Since the final voltage drop across the capacitors is 10V, what is the final value of a?

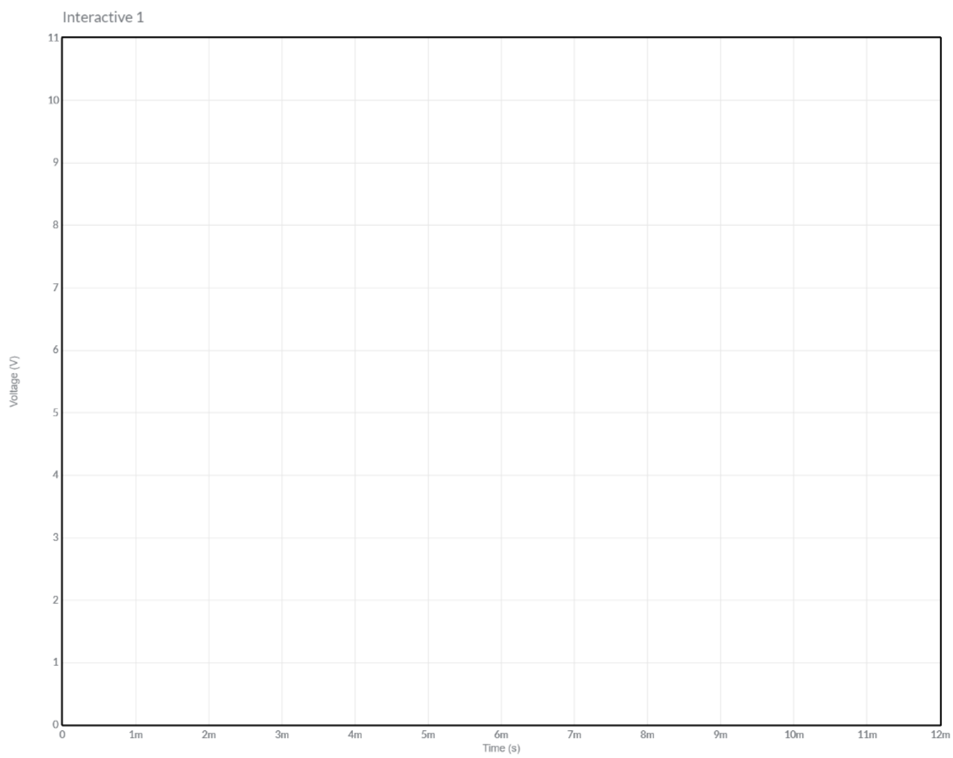
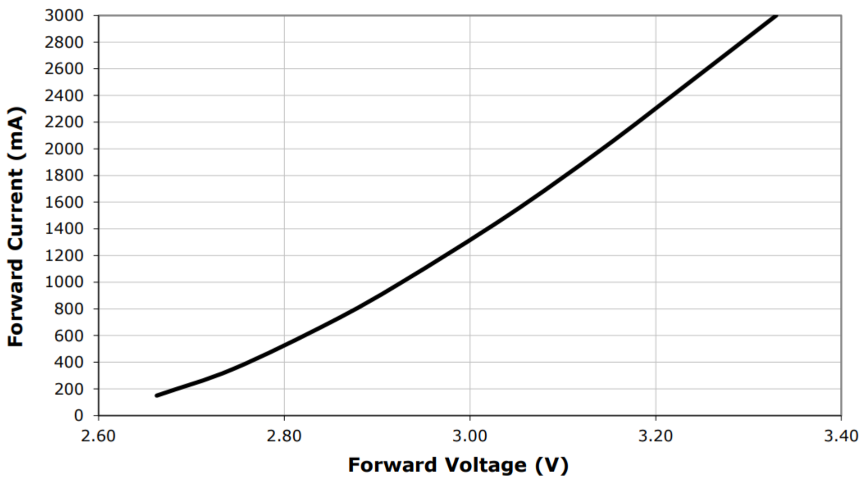


*Figure 2: Plot the voltage at node **a** for the circuits shown in 001 and 002*

Practical Application – Power dissipated by BJT

You are required to illuminate a Cree Xlamp XM-L2 High Power LED using a current source. The LED has a I vs. V curve is shown at left in Figure 1. Your current source, shown at right in Figure 1, uses a TIP31 (a high powered NPN transistor) and a 2.7V Zener diode. You need to run the LED at 2 amps.



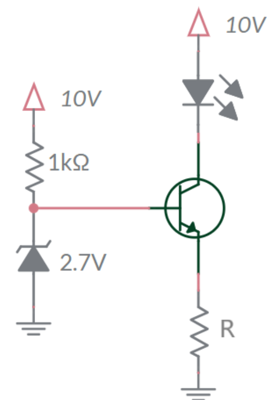


Figure 3: (Left) I/V curve for a Cree Xlamp XM-L2 High Power LED. (Right) The constant current source used to drive the LED.

a. What is the emitter voltage?

b. What value of emitter resistor, R, will produce the needed 2A through the LED?

c. What is the voltage drop across the LED at 2A?

d. What is the power dissipated by the LED at 2A?

e. What is the power dissipated by the resistor R at 2A?

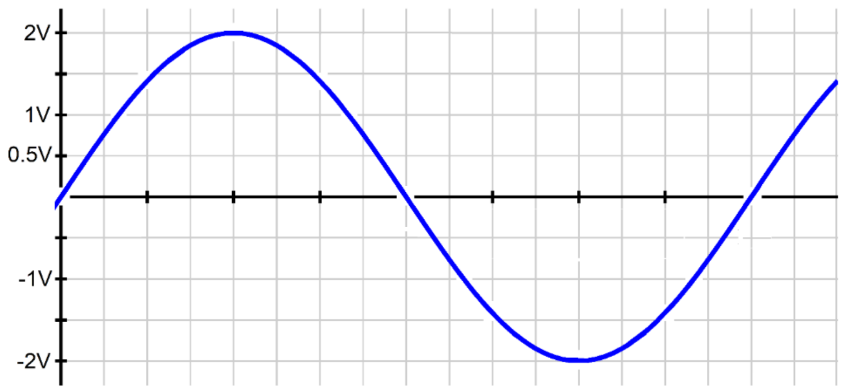
f. What is the voltage across the BJT at 2A?

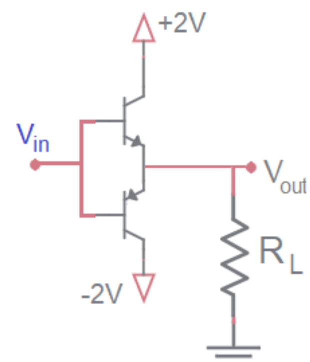
g. What is the power dissipated by the BJT at 2A?

h. How can you reduce the power dissipation of the BJT while keeping 2A running through the LED? Do not change the circuit topology.

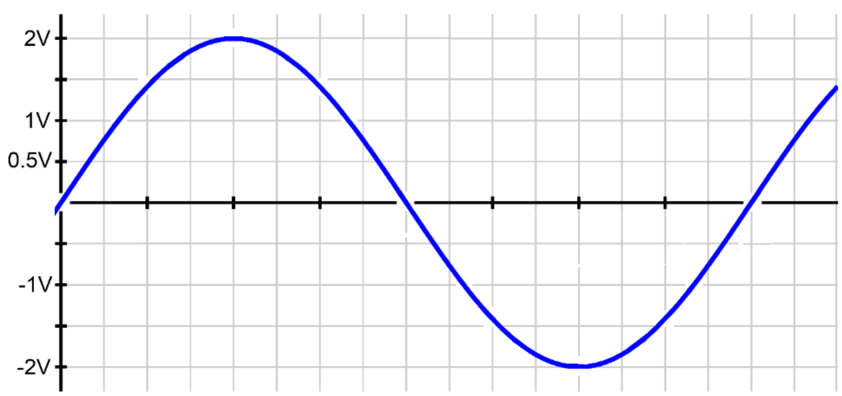
Practical Application - Cross Over Distortion

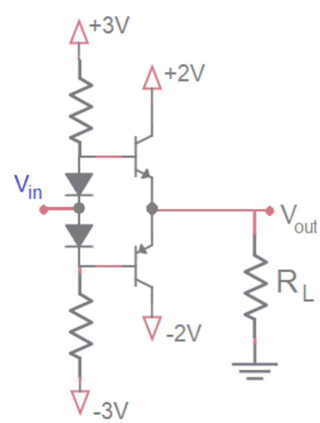
The blue sine wave shown in the graph below is applied as the input, V_{in}, to the circuit shown below at right. Determine the output, V_{out}, and graph it on the graph below. The value of the load resistor, R_L is unimportant.



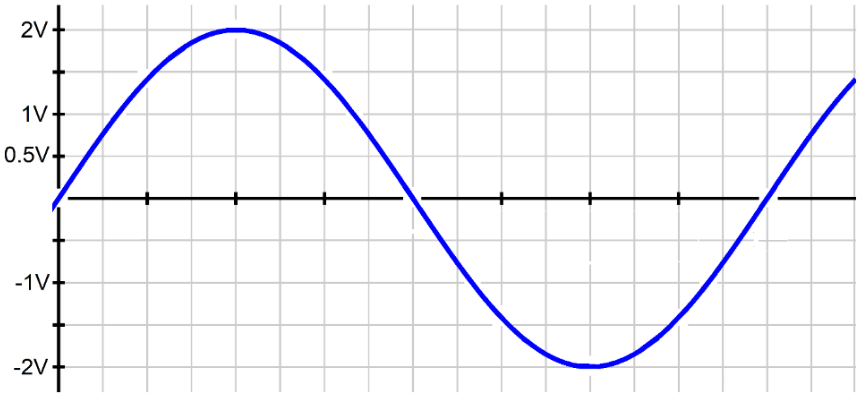


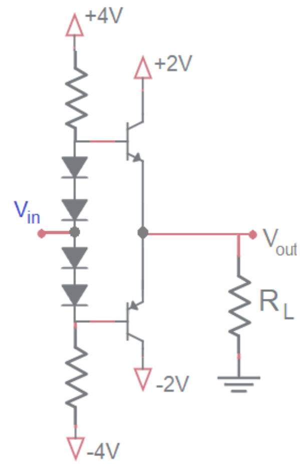
The blue sine wave shown in the graph below is applied as the input, V_{in}, to the circuit shown below at right. Determine the output, V_{out}, and graph it on the graph below. The value of the resistors are unimportant.





The blue sine wave shown in the graph below is applied as the input, V_{in}, to the circuit shown below at right. Determine the output, V_{out}, and graph it on the graph below. The value of the resistors are unimportant.



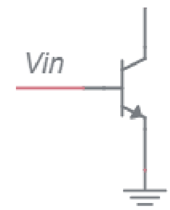


Practical Application – Negative feedback by emitter degeneration

In the Ebers-Moll equation of a BJTs the collector current is described as

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

The I_S term, the reverse saturation current, is strongly temperature dependent. According to multiple sources, it doubles for every 10°C rise in temperature. This dependency can lead to thermal runaway in the common emitter circuit show at right. In the following questions assume that Vin is constant and that the BJT is operating in the active region.



1. Look at the schematic at right, does the base emitter voltage depend on the collector current?

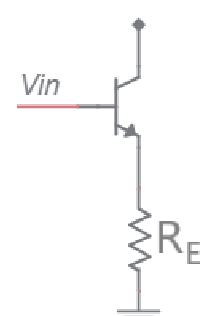
2. If the temperature of the BJT increases, does I_s increase or decrease?

3. If I_S increases, does I_C increase or decrease?

4. If I_C increases does the power dissipated by the BJT increase or decrease? Assume V_C is constant.

5. If the power dissipated by the BJT increases, does the temperature of the BJT increase or decrease?

From these inferences you should see that an increase in the BJT's temperature causes further increase in the temperature of the BJT – thermal runaway. This is an example of positive feedback – a tendency of a system to increase a parameter without bound. While you may like positive feedback, electrical systems can do bad things when they have too much positive feedback.



Now consider the circuit shown at right. This is a common emitter with emitter degeneration resistor. As we will see, the addition of a resistor to the emitter adds negative feedback which greatly decreases the tendency of the BJT to experience thermal runaway. In the following questions assume that Vin is constant and that the BJT is operating in the active region.

1. Look at the schematic at right. Does the base emitter voltage increase or decrease when the emitter current increases?

2. When the collector current increases, does the emitter current increase or decrease?

3. If the temperature of the BJT increases, does I_s increase or decrease?

4. If I_S increases, does I_C increase or decrease?

5. If I_C increases, does I_E increase or decrease?

6. If I_E increases, does V_E increase or decrease?

7. If V_E increases, does V_{BE} increase or decrease? Assume that V_B is being held constant.

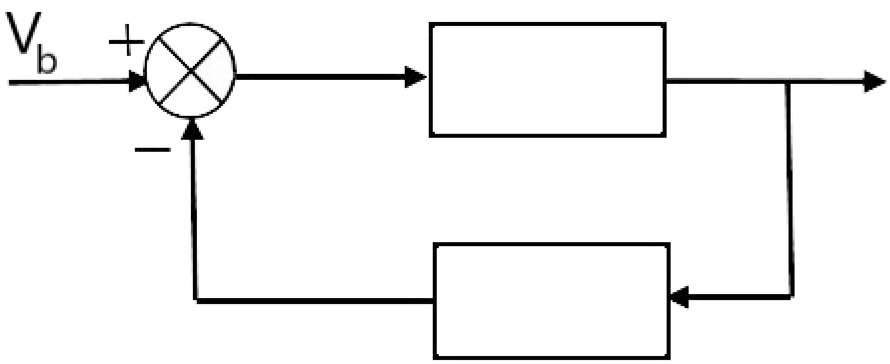
8. If V_{BE} decreases, does I_C increase or decrease?

9. If I_C decreases, does the power dissipated by the BJT increase or decrease?

10. If the power dissipated by the BJT decreases, does the temperature of the BJT increase or decrease?

The ability of the deceasing V_{BE} being able to keep the increasing I_S in check depends on the magnitude of the emitter resistor. If it the emitter resistor is too small, then the BJT may still exhibit thermal runaway. In that case, you should increase the emitter resistor until thermal runaway is kept in check.

Now, let's see if you can make a feedback diagram to describe this negative feedback using the diagram below. Annotate the diagram according to the instructions below.

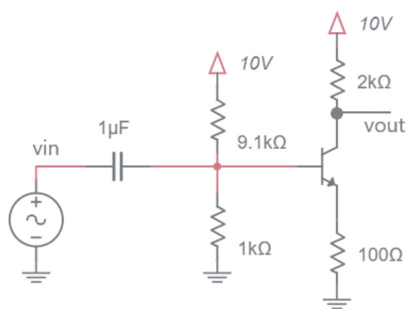


1. Make the reference input equal to V_b (this is the input at the left).
2. What must the inverting input to the summer junction be for the output of the summer junction to be V_{be}? Write this signal and V_{be} on the diagram.
3. Fill in the upper block with the right-hand side of the Ebers-Moll equation.
4. Label the output of the block diagram with the left-hand side of the Ebers-Moll equation. This should make sense because the Ebers-Moll equation is a bunch of constants with two variables.
5. Fill in the lower block with an equation which converts the block diagram output into the variable used in the inverting input to the summer junction. Hint, its Ohm's law.

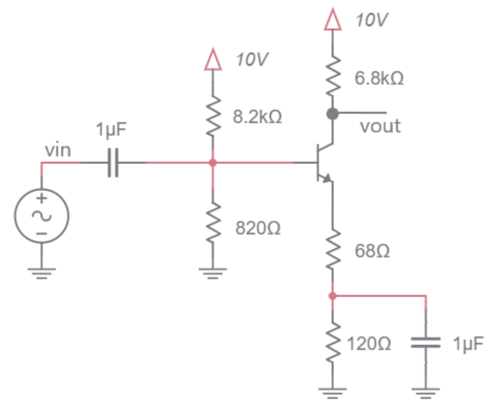
Chapter: BJT @ AC

Small signal analysis of BJT circuits

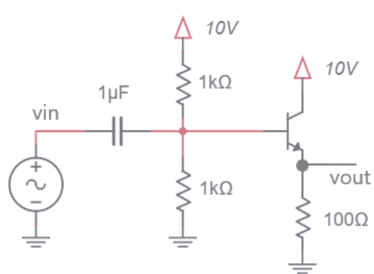
- In the I_C and V_{out} columns enter the DC collector current and DC output voltage respectively.
- In the other columns, enter the small signal values.
- Assume an infinite Early voltage.
- On separate page(s), show your work for each problem.
- Round you answer to 2-significant figures and include units.



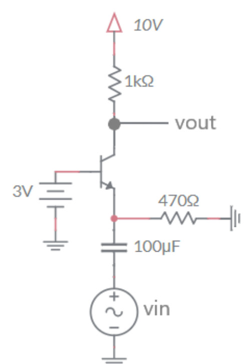
001



002



003

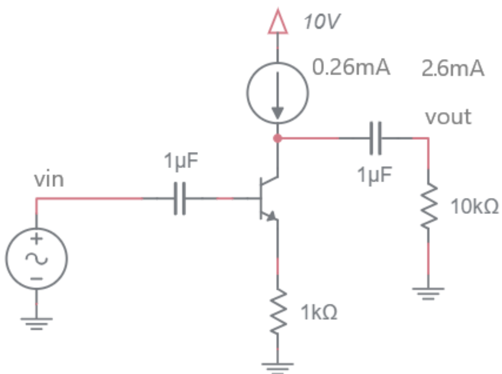


004

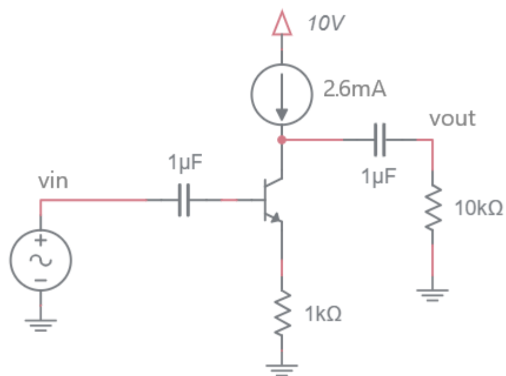
Circuit	I_C	V_{OUT}	g_m	r_e	A_V
001					
002					
003					
004					

Small signal analysis of BJT circuits

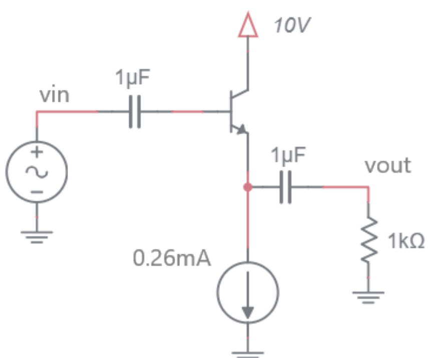
- In the I_C and V_{out} columns enter the DC collector current and DC output voltage respectively.
- In the other columns, enter the small signal values.
- Assume an infinite Early voltage.
- On separate page(s), show your work for each problem.
- Round you answer to 2-significant figures and include units.



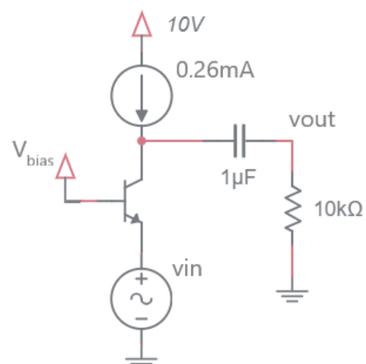
001



002



003

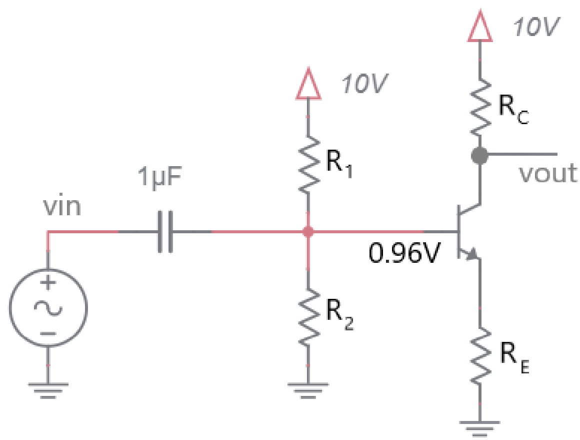


004

Circuit	I_C	V_{OUT}	g_m	r_e	A_V
001					
002					
003					
004					

Small signal analysis of BJT circuits

Let's explore the range of AC gain that you can achieve with the following circuit. The values of R1 and R2 have been selected so that the base of the BJT is DC biased to 0.96V – you do not need to compute the resistors values needed to make this happen.



Step 1: Describe I_C in terms of R_E

Step 2: Describe V_{OUT} in terms of R_E and R_C

Step 3: Describe g_m and r_e in terms of R_E

Step 4: Draw a small signal model for the circuit

Step 5: Describe the small signal gain in terms of R_E and R_C

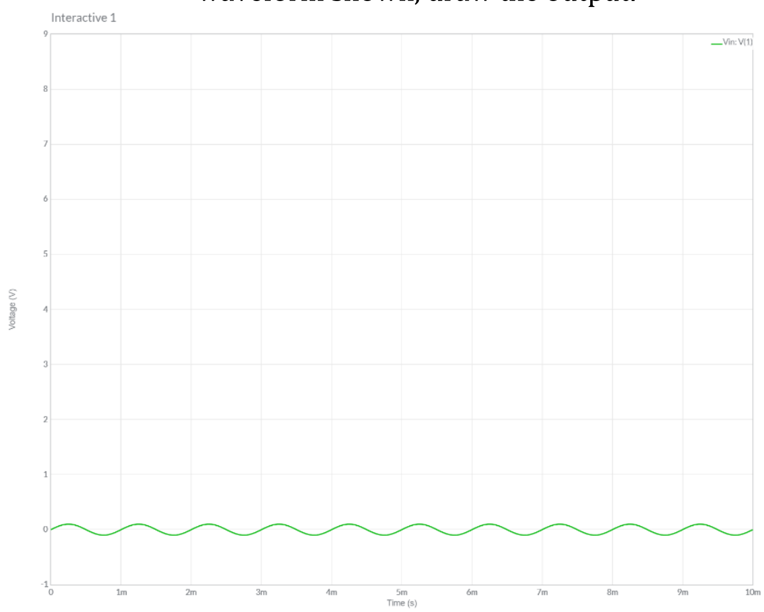
Step 6: Describe V_{OUT} in terms of A_v

Step 7: Find the maximum gain before the BJT saturates

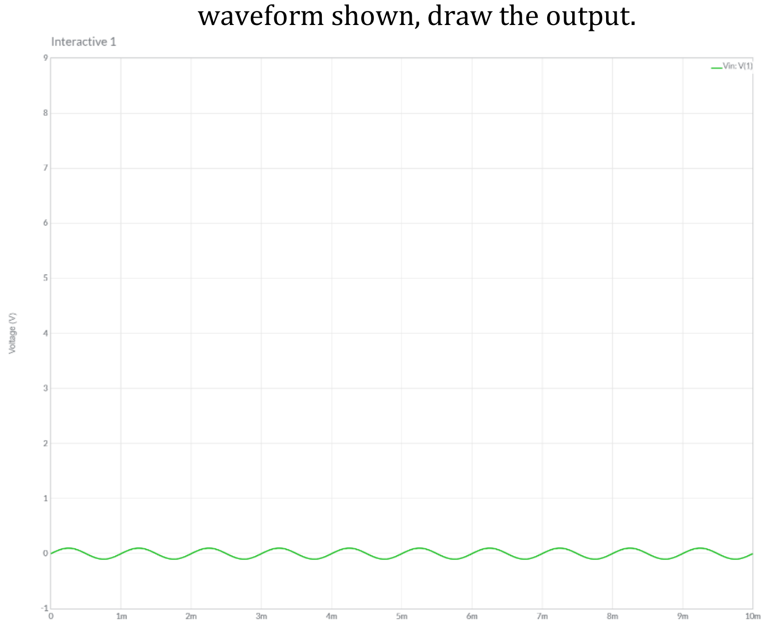
Examining the input/output behavior of various amplifiers

In both of the following problems the BJT amplifier is **powered by 10V and GND.** The input is capacitively coupled to the amplifier (there is no issue with the input swinging to negative voltages).

a. You have built a BJT amplifier where your DC analysis of the circuit shows that Vout sits at 5V. Your AC analysis shows that your circuit has a gain of -2. Given the input waveform shown, draw the output.



b. You have built a BJT amplifier where your DC analysis of the circuit shows that Vout sits at 5V. Your AC analysis shows that your circuit has a gain of 6. Given the input waveform shown, draw the output.



Chapter: Frequency Domain

Apply definitions of decibels and decades

1. (vin/vout to dB) If an electronic component takes a 1.0v sine wave as input and output a 0.1v sine wave, how much attenuation would it have?

2. (vin/dB to vout) If you input a 2.4 V sin wave into a circuit that attenuated it by -32dB, what would the amplitude of the output waveform be?

3. (out/dB to vin) A circuit attenuates an input signal by -68 dB to produce a 0.5 V signal. What is the amplitude of the input signal?

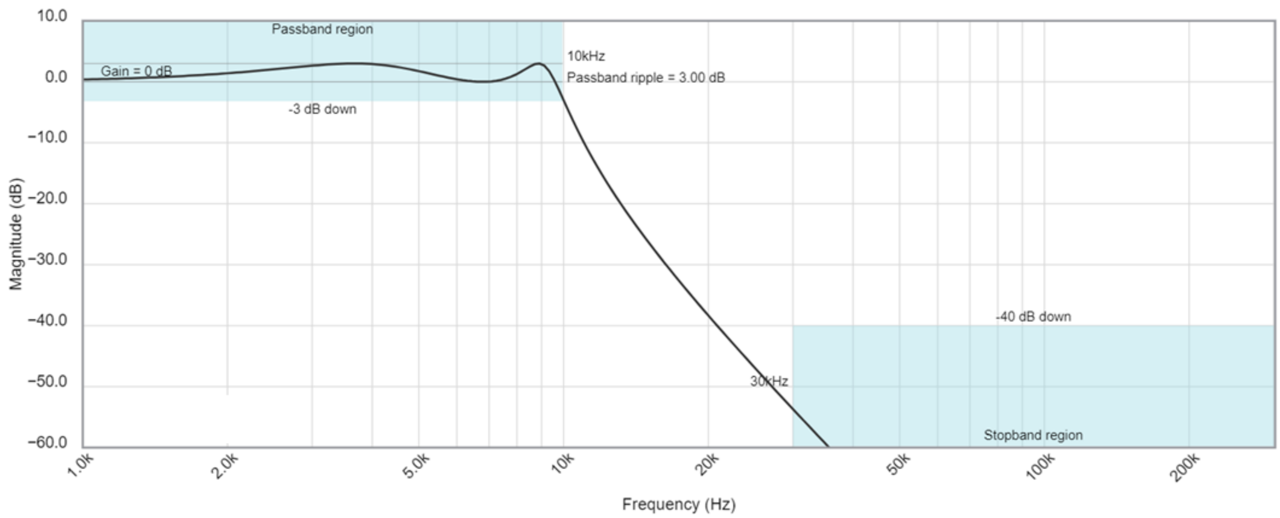
4. (fhi/flo to decades) How many decades separate 10kHz and 30kHz?

5. (fhi/decades to flo) What frequency is 1.5 decades below 300kHz?

6. (flo/decades to fhi) What frequency is 1.5 decades above 10kHz?

Interpret a real Bode plot

The following Bode plot is from a Chebyshev low pass filter with corner frequency of 10kHz. Use the information in the Bode plot to answer the following questions in the boxes provided.

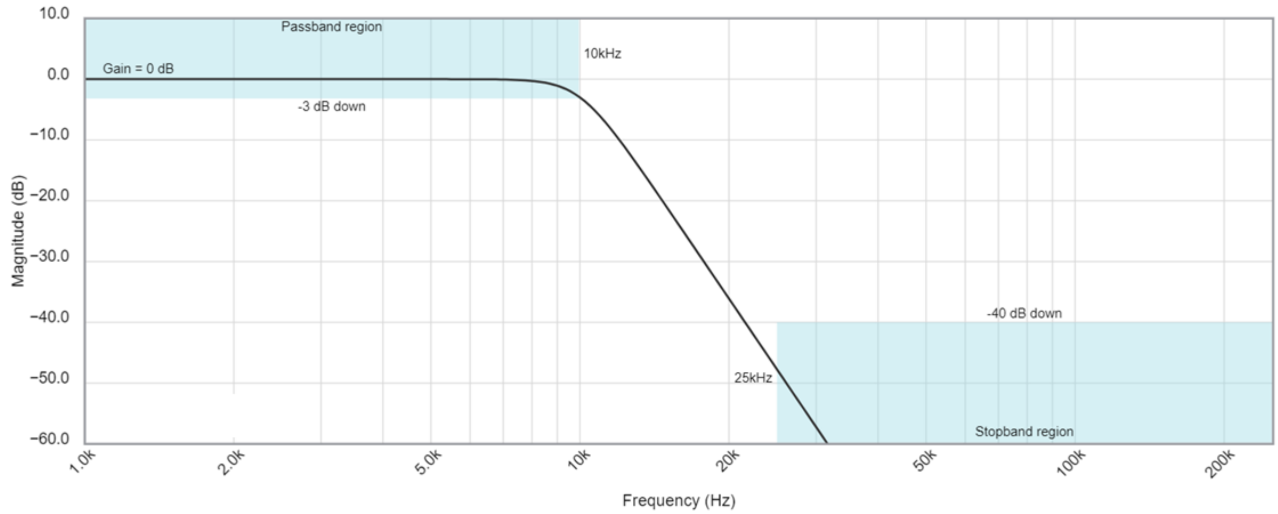


7. If you apply a 20kHz sin wave to this filter, how much attenuation will the output experience?

8. You want -45dB of attenuation, what frequency sin wave should you apply?

9. Estimate the slope of the magnitude in the stop band and use this to estimate the order of the filter. Hint, use the corner frequency as one of your points.

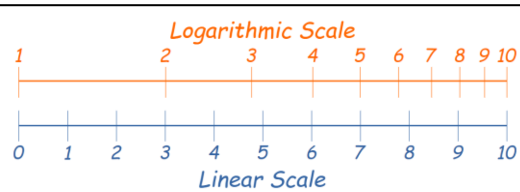
The following Bode plot is from a Butterworth low pass filter with corner frequency of 10kHz. Use the information in the Bode plot to answer the following questions in the boxes provided.



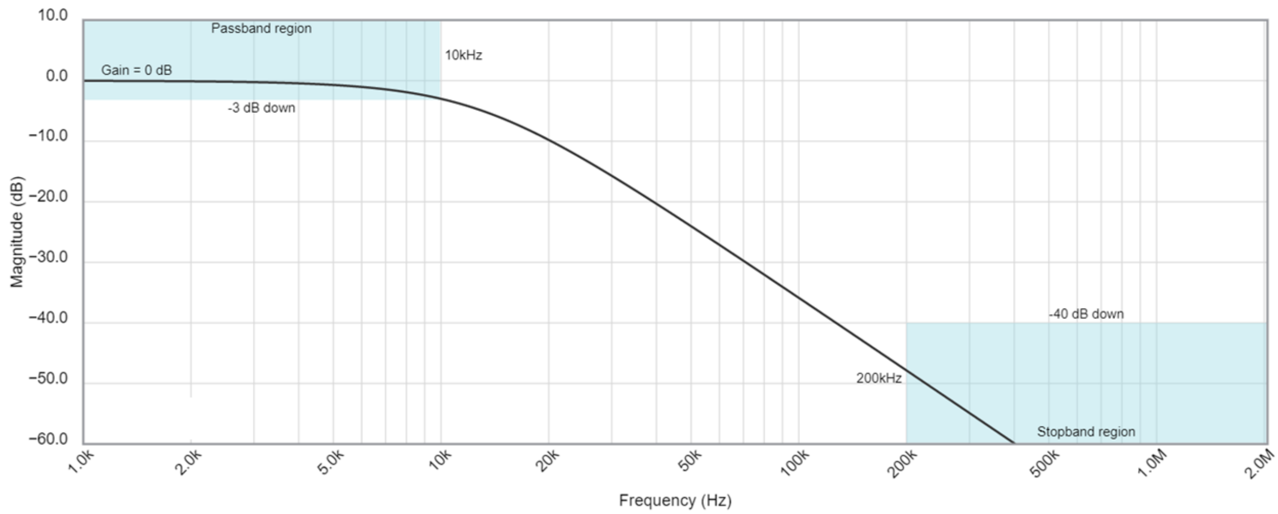
10. If you apply a 25kHz sin wave to this filter, how much attenuation will the output experience?

11. You want -35dB of attenuation, what frequency sin wave should you apply?

12. Estimate the slope of the magnitude in the stop band and use this to estimate the order of the filter. Hint, use the corner frequency as one of your points.



The following Bode plot is from a Bessel low pass filter with corner frequency of 10kHz. Use the information in the Bode plot to answer the following questions in the boxes provided.



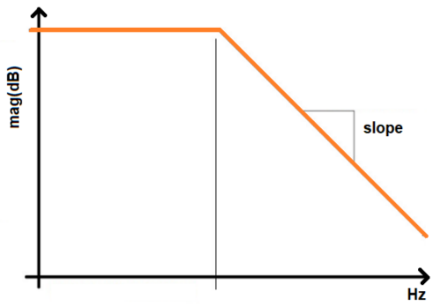
13. If you apply a 20kHz sin wave to this filter, how much attenuation will the output experience?

14. You want -55dB of attenuation, what frequency sin wave should you apply?

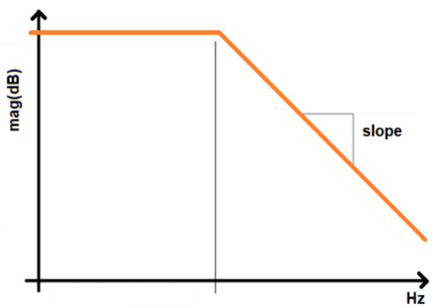
15. Estimate the slope of the magnitude in the stop band and use this to estimate the order of the filter. Hint, use the corner frequency as one of your points.

Interpret an ideal Bode plot

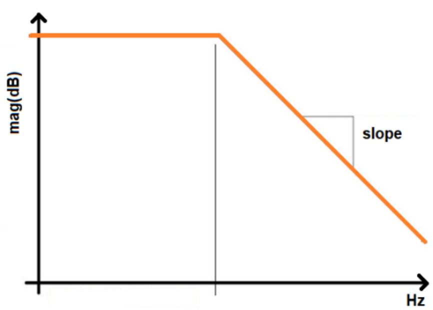
1. Determine the output attenuation (in dB) from a 80kHz input signal from a 1st order LPF with a corner frequency of 2kHz that has 0dB of attenuation in the passband.



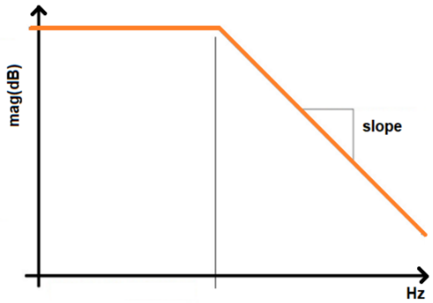
2. Determine the input frequency needed to attenuate an input signal by -48dB from a 1st order LPF with a corner frequency of 2kHz that has 0dB of attenuation in the passband.



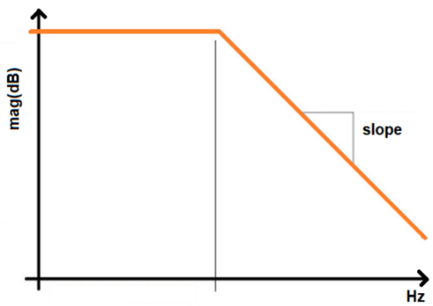
3. Determine the corner frequency of a 4th order LPF that has 0dB of attenuation in the passband and attenuates a 130kHz input by -68dB.



4. Determine the order of a filter that has 0dB of attenuation in the passband, a 15kHz corner frequency and attenuates a 90kHz input by -77.8dB.



5. Determine the magnitude in the passband of a second order LPF, an 8kHz corner frequency and attenuates a 56kHz input by -13.8dB.



Create a Bode plot from pole/zero map

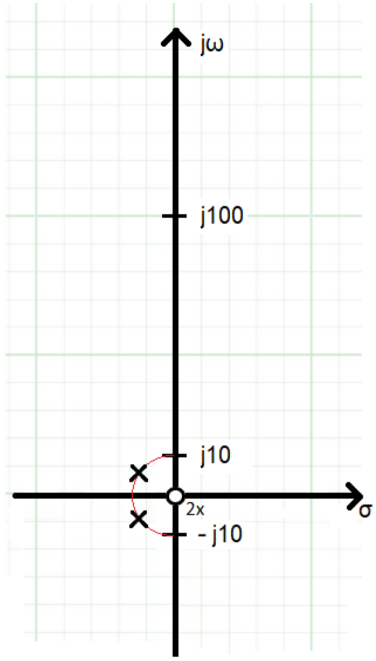
Use the pole/zero plots for the following problems to draw the Bode plots. You should:

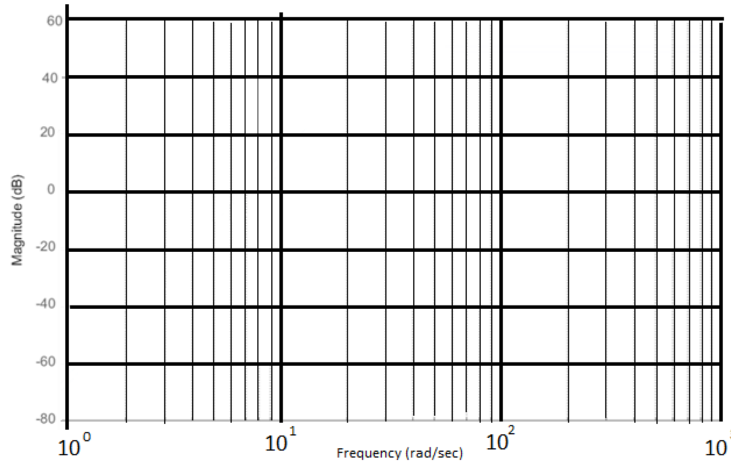
- Annotate the slope values (dB/decade) and draw them correctly on the Bode plot.
- Use the correct corner frequencies.
- In the absence of compelling evidence, assume a 0dB passband.
- If not shown, the complex conjugate poles/zeros do not matter.

You are welcome to check your answers using Matlab – this is much easier than you might. Here is some starter code. Note that Matlab will add lots of details that you are not expected to show in your rough plots.

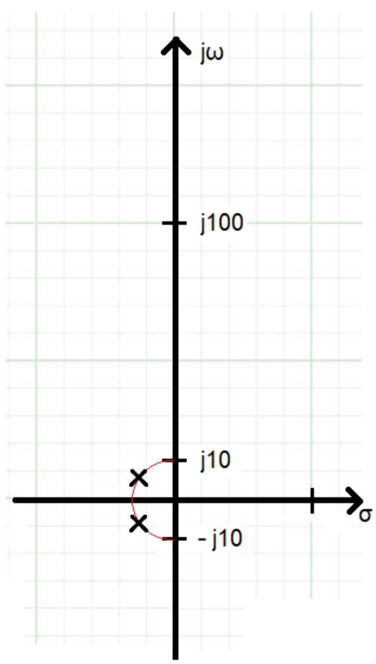
```
s = tf('s');            % define the complex variable s
p1 = 100j - 1;          % define the coordinates of a pole
z1 = 100j + 1;          % define the coordinates of a zero
G = (s-z1)/(s-p1);      % define the transfer function that has this constellation
pzmap(G);               % Show the pole zero map
waitforbuttonpress();   % waits for a mouse click on the plot to move on
bodeplot(G);            % plot it!
```

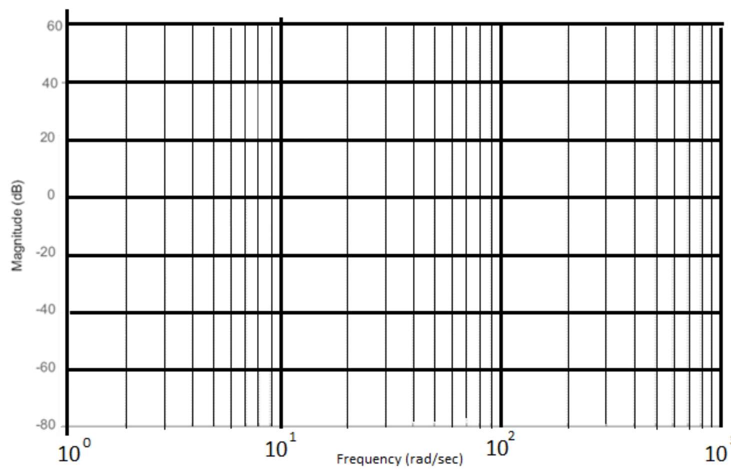
1. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. There are two zeros at the origin. The semicircle has a radius of 10.



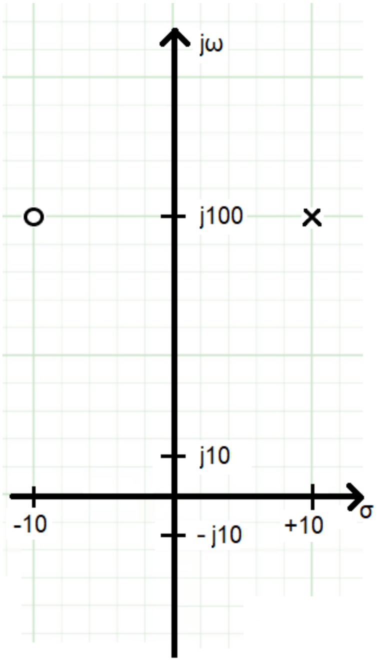


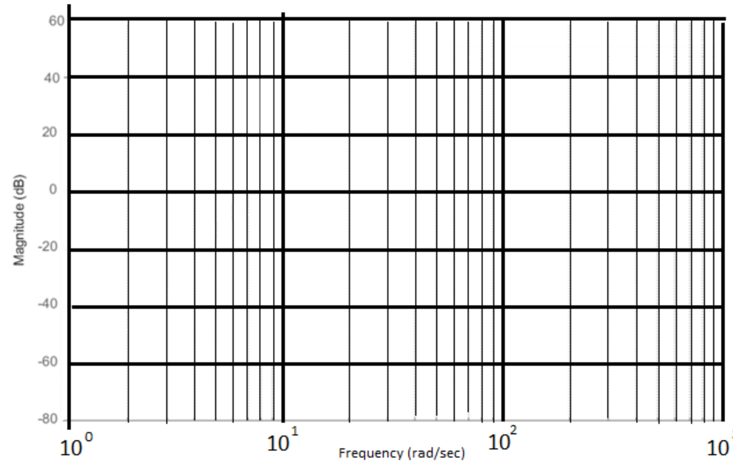
2. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The semicircle has a radius of 10.



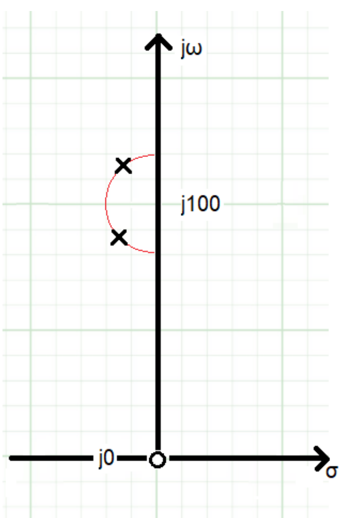


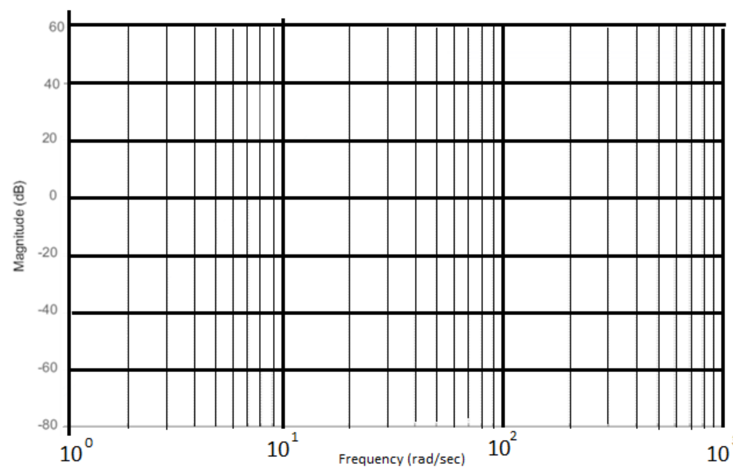
3. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently.



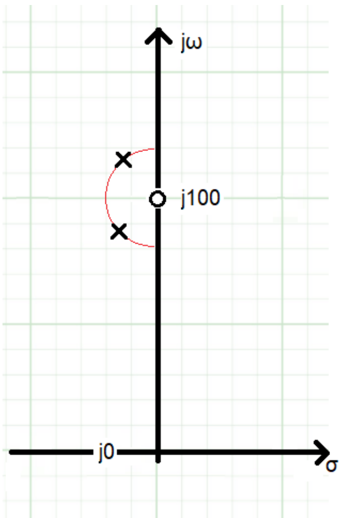


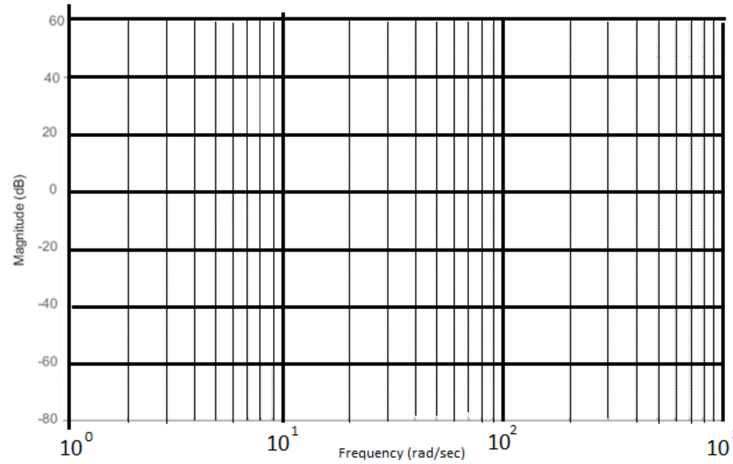
4. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. There is one zero at the origin. The semicircle has a radius of 10.



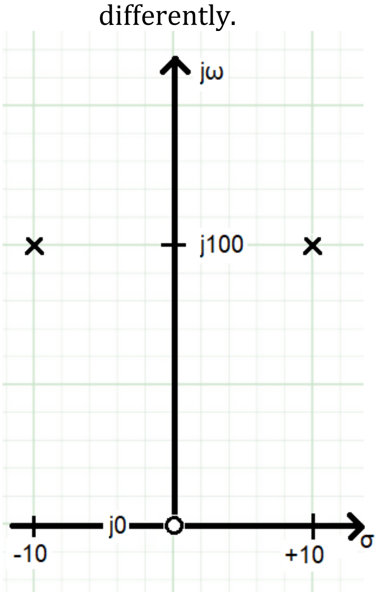


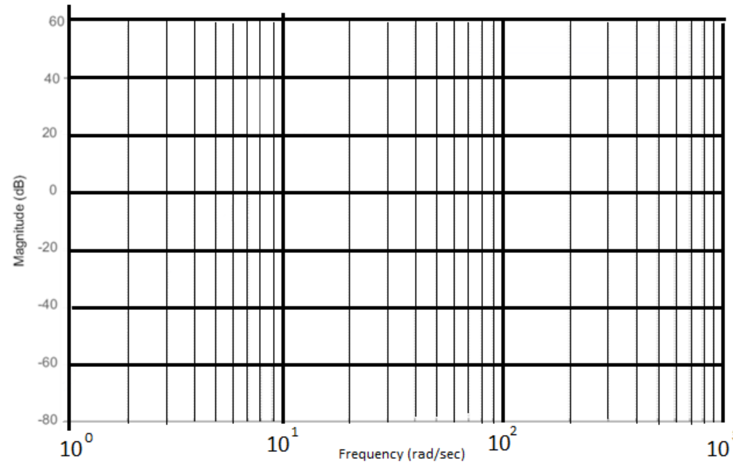
5. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. There is one zero at j100. The semicircle has a radius of 10.





6. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently.





Gain product bandwidth of an op amp

1. You are working with an opamp that has a 1MHz gain-bandwidth product. Your circuit is configured to have a gain of 30dB.

    a. What is the bandwidth of your circuit?

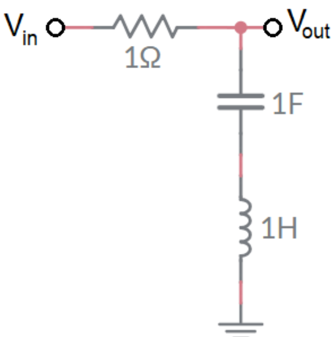
    b. What is the gain (in dB) of a sinusoidal input with a frequency of 10kHz?

    c. What is the gain (in dB) of a sinusoidal input with a frequency of 100kHz?

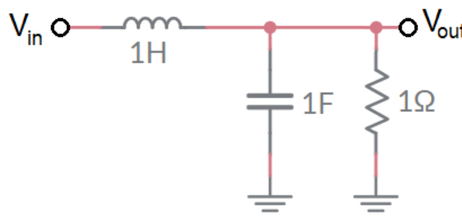
    d. At what frequency will the gain drop to 25dB?

Deriving Transfer Functions

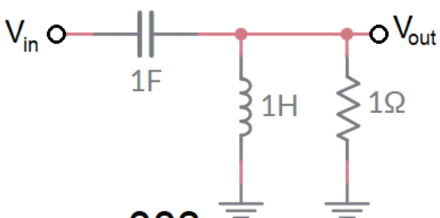
Determine the transfer function for the following circuits. Write your transfer function as the product of constants, integrators and canonical first and second order transfer functions (shown in Table 1). In the Type column indicate if the transfer function describes a low pass filter, high pass filter, band pass filter, or notch filter. You are welcome to use to simulator to verify.



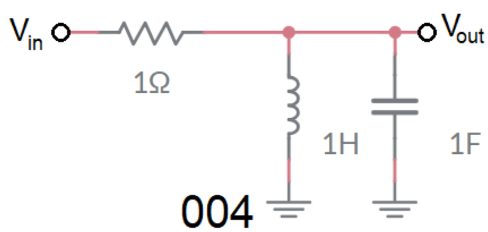
001



002



003



004

	Transfer function	Type
001		
002		
003		
004		

Practical Application – Buck Converter Filter

A buck converter consists of two subcircuits, a square wave generator and a filter shown in Figure 4. The square wave generator is built from the MOSFET Q3 (represented as a switch). The opening and closing of this switch generated voltage pulses on Vin. The filter consists of the inductor, capacitor and resistor.

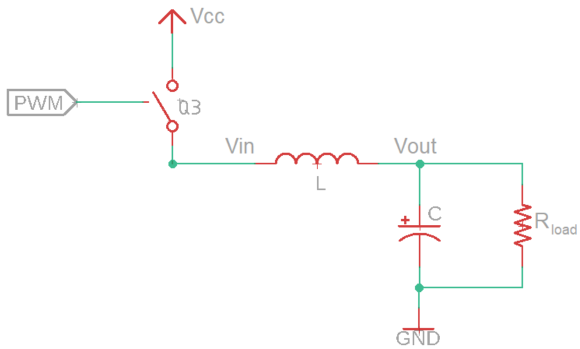


Figure 4: The essential components of a buck converter.

Derive the transfer function Vout(s)/Vin(s) for the filter. Hint, you may want to use voltage divider. Put your answer in canonical form.

What type and order filter is this?

Substitute L=1mH, R = 100Ω, and C = 100uF into your transfer function. Use the definitions of canonical form below to determine the corner frequency of the filter.

Table 1: Canonical form for first and second order systems.

	First Order	Second Order
Form	$k\dfrac{\alpha}{s+\alpha}$	$k\dfrac{\omega_n^2}{s^2+2\zeta\omega_n+\omega_n^2}$
Corner Freq	α rad/sec	ω_n rad/sec

What is the corner frequency of this filter?

Why does this filter output the average value of a 10kHz PWM waveform? Hint, construct the Fourier series of the 10kHz PWM wave, then consider which harmonics the filter passes and which it blocks.

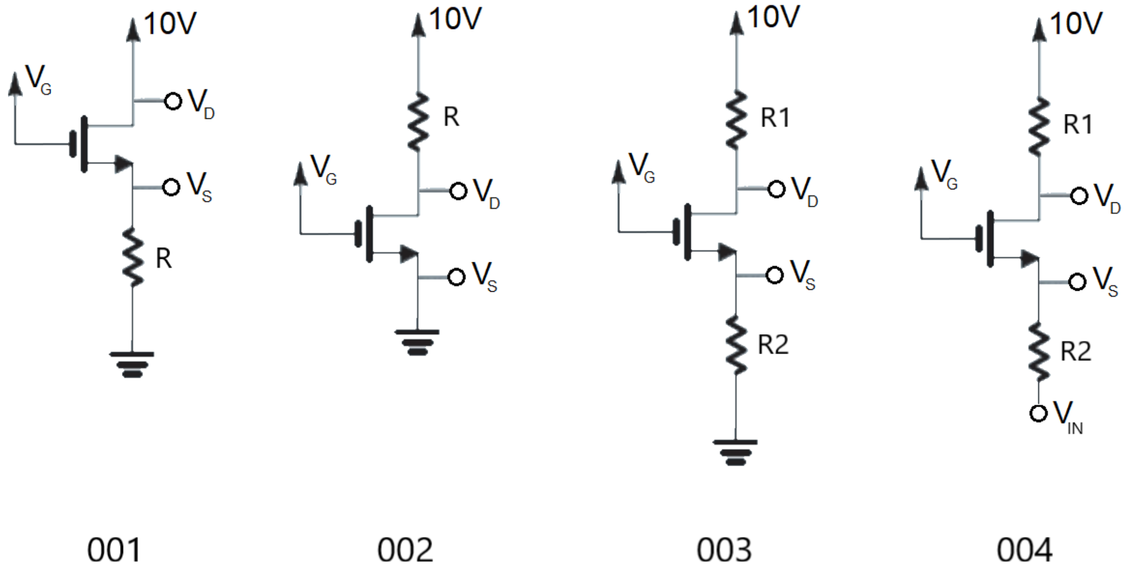
Chapter: MOSFETs

MOSFET circuits with Resistor(s) and DC source.

Fill in the blanks in the following table given the four circuit configurations below. Assume V_T = 1V and k = 1mA/V^2. In the Mode column enter one of the following:

- "Off" when $V_G<V_T$,
- "Triode" when $V_{DS} < V_{GS} - V_T$ or
- "Saturation" when $V_{DS} >= V_{GS} - V_T$.

When the MOSFET is in the triode mode put "??" in the I_D column and any column that depends on I_D. When the MOSFET is in off mode, put "??" in the I_D column and any column that depends on I_D.



Circuit	V_D	V_G	V_S	V_{GS}	V_{OV}	R	I_D	Mode
001		0V				xx		
001		3V				1k		
001		6V				1k		
001		3V				10k		
001		6V				10k		
001		3V				100k		

Circuit	V_D	V_G	V_S	V_{GS}	V_{OV}	R	I_D	Mode
002		0V				x		
002		3V				100Ω		
002		3V				1kΩ		
002		3V				10kΩ		
002		6V				100Ω		
002		6V				1kΩ		

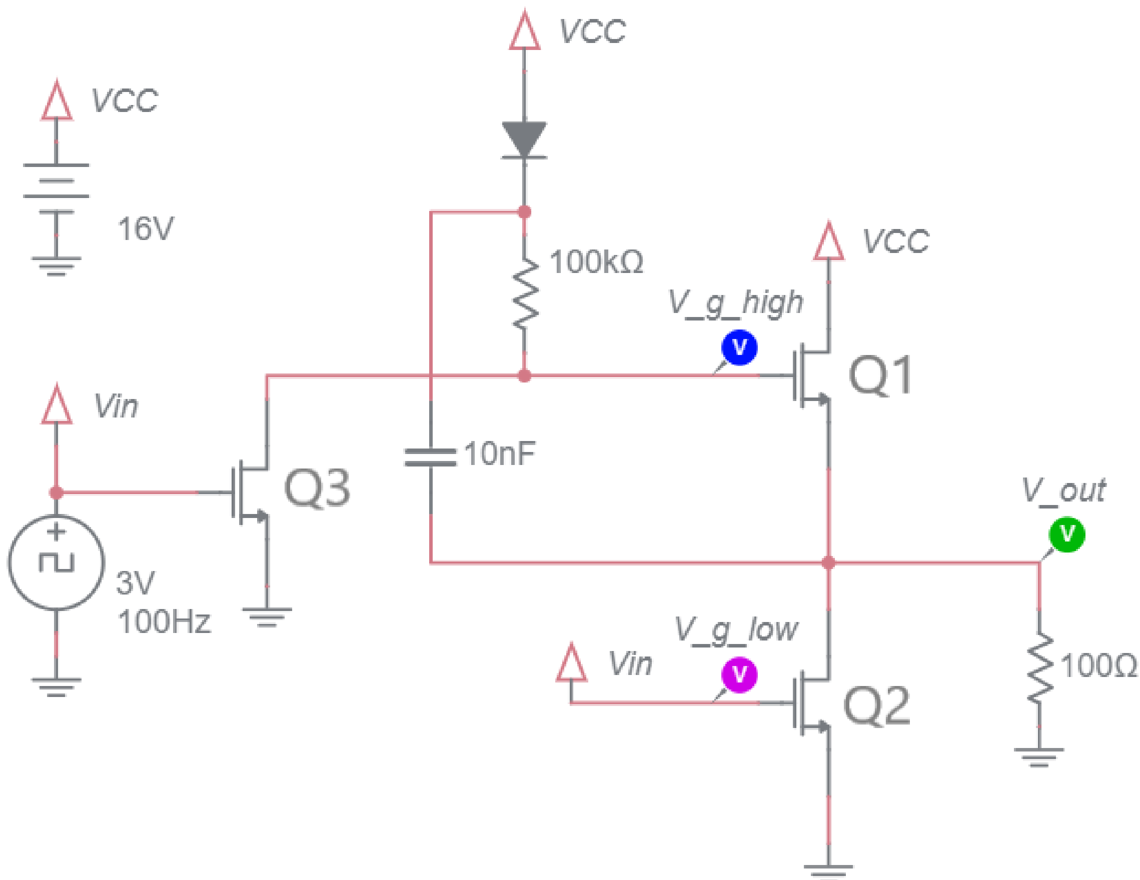
Circuit	V_D	V_G	V_S	V_{GS}	V_{OV}	R1	R2	I_D	Mode
003		0V				xx			
003		3V				1kΩ	1k		
003		6V				1kΩ	1k		
003		3V				10k	10k		
003		3V				100k	1k		
003		6V				10k	1k		

Let Vin = 3V

Circuit	V_D	V_G	V_S	V_{GS}	V_{OV}	R1	R2	I_D	Mode
004		6V				1kΩ	1kΩ		

Practical Application - Half Bridge circuit with bootstrap capacitor

The following circuit allows a low power signal, modeled as the 3V, 100Hz square wave in the schematic, to control a high voltage, high power load, modeled as the 100Ω resistor. In order to understand how the following circuit accomplishes this, work through the following questions.



Assume that Vin is 3V

1. Do MOSFETs Q2 and Q3 act more like open switches (allowing no current to flow) or closed switches (allowing current to flow)?
2. Assume MOSFET Q3 is acting like a closed switch
    a. What is the gate voltage of MOSFET Q1?
    b. Can V_{gs} of Q1 ever be greater than V_T?
    c. Does Q1 act like an open switch or closed switch?

The switch settings you have discovered are incorporated in the schematic to produce the modified schematic shown below left.

3. After a short delay, what will be the voltage across the 10nF capacitor. For simplicity, assume an ideal diode model.

Assume that Vin is 0V

1. Do MOSFETs Q2 and Q3 act more like open switches (allowing no current to flow) or closed switches (allowing current to flow)?
2. Assume MOSFET Q3 is acting like an open switch and that the 10nF is fully charge from a period where Vin was 3V
    a. What is V_{gs} for MOSFET Q1? Hint the answer is dictated by the voltage stored on the 10nF capacitor.
    b. Does Q1 act like an open switch or closed switch?

   The switch settings you have discovered are incorporated in the schematic to produce the modified schematic shown below right.

3. The charge on the top plate of the capacitor flow could flow
    a. Through the 100kΩ and on to the drain of Q3 to ground
    b. Through the 100kΩ and on to the gate of Q1
    c. Through the diode to Vcc

   In each case, explain why no charge will flow off the top plate of the capacitor.

MOSFET behavior when Vin = 3V	MOSFET behavior when Vin = 0V